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(54) **DECENTRALIZED POWER EXCHANGE**

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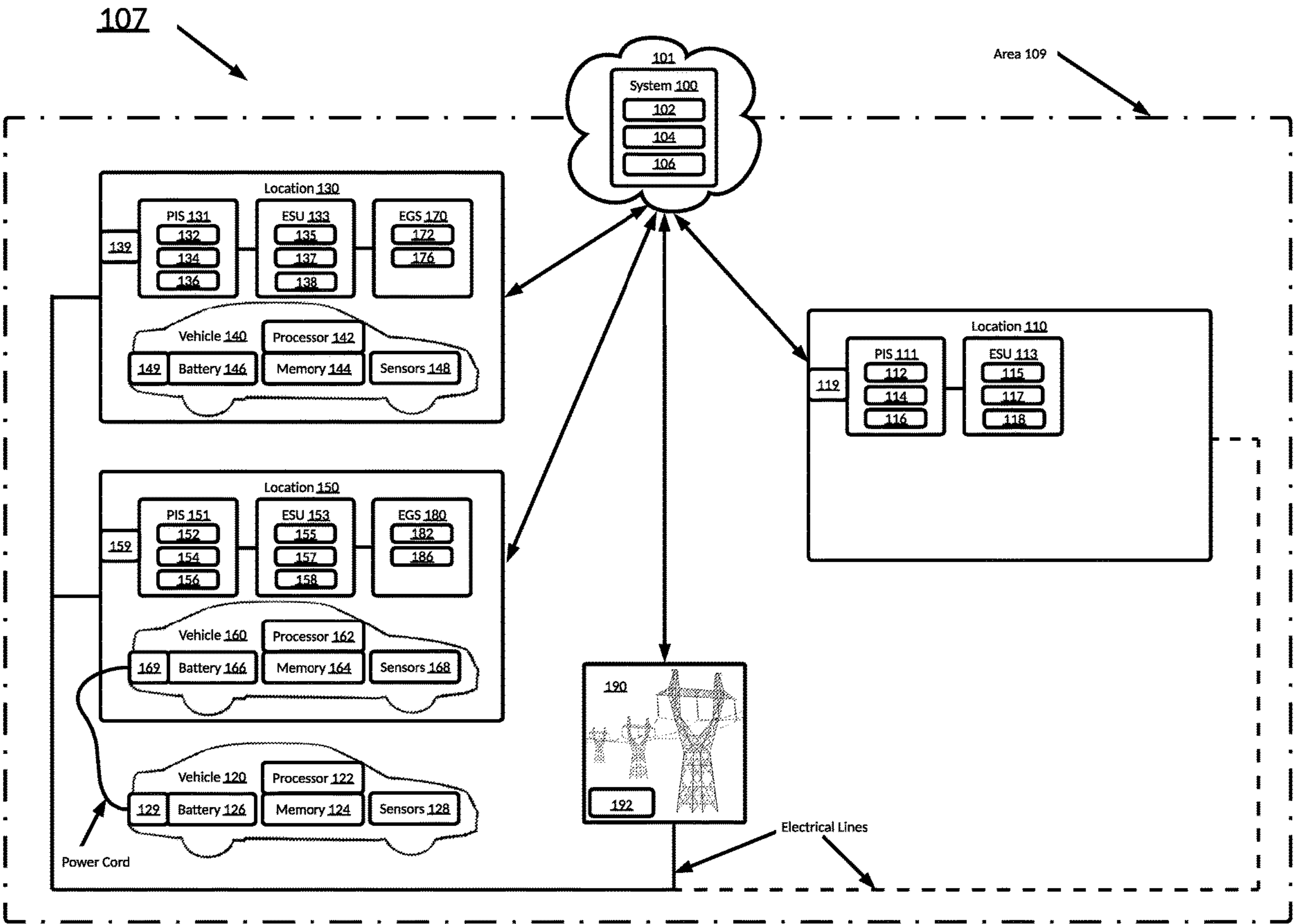
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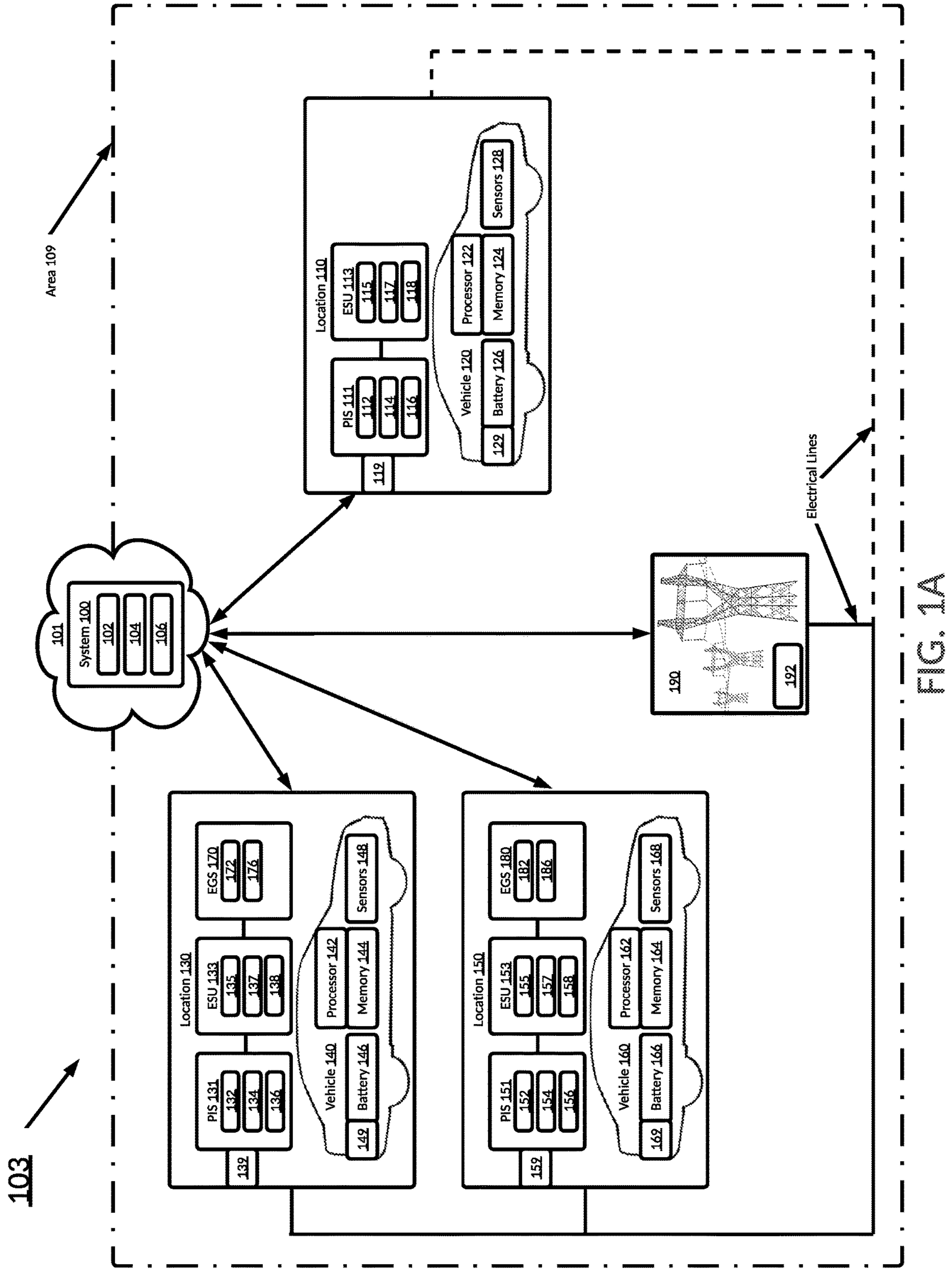
(57) **ABSTRACT**

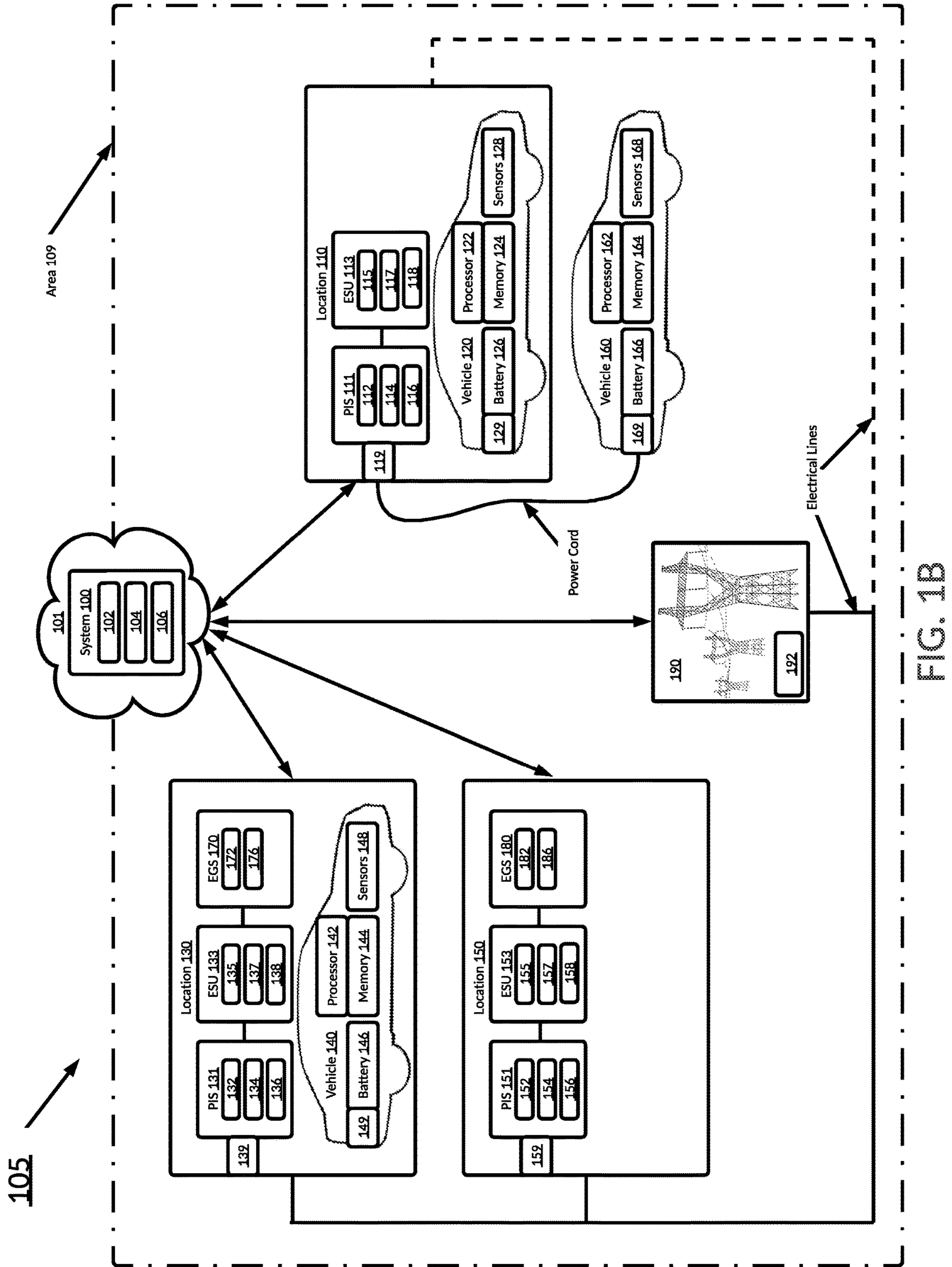
An example operation includes one or more of determining an inability for an entity to provide electricity over a threshold to an area; determining available energy from a group of electric vehicles from locations currently receiving energy in the area and a group of energy storage units in the area greater than the threshold; providing the determined available energy to locations not currently receiving energy from the entity.

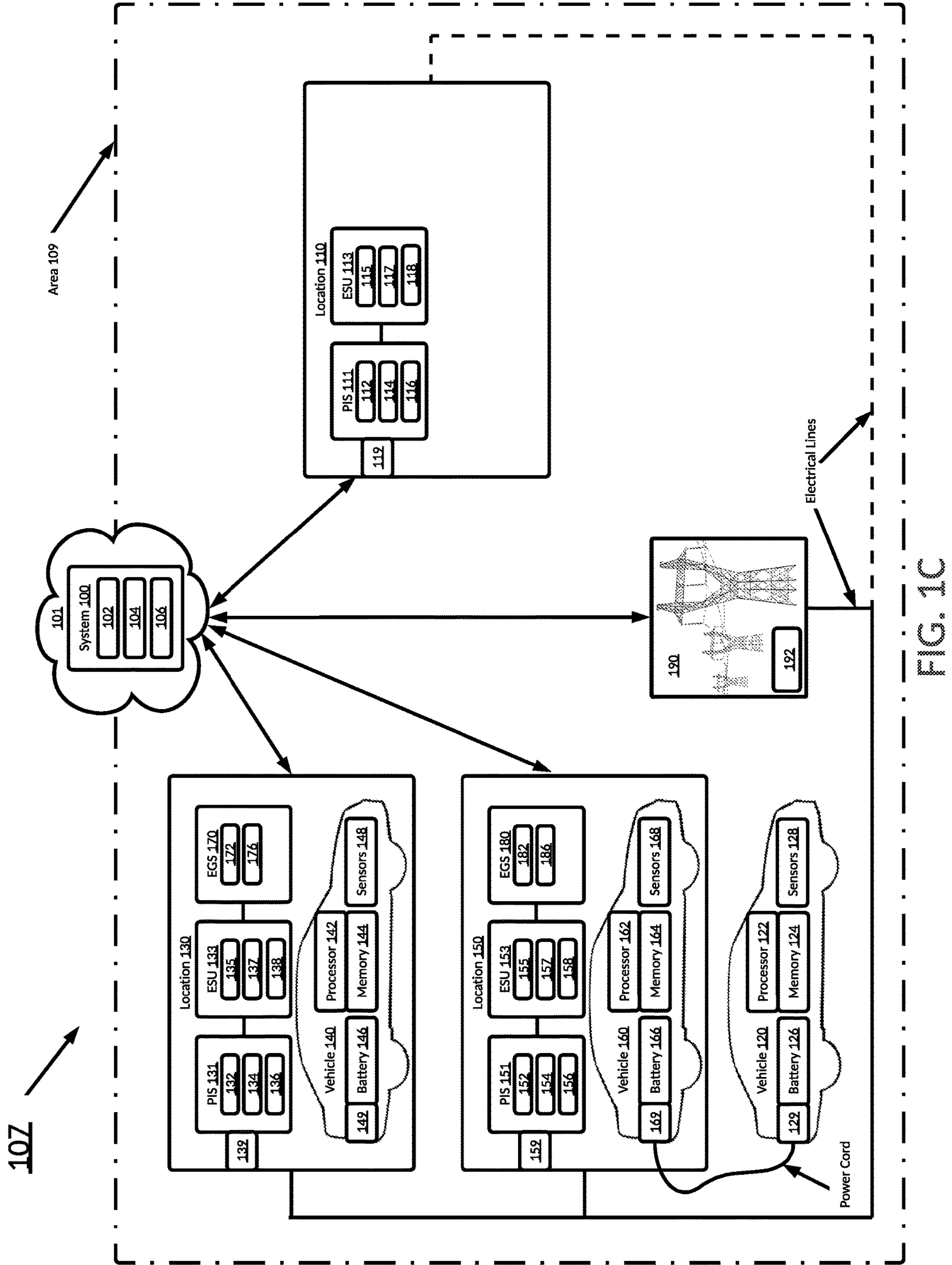
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200

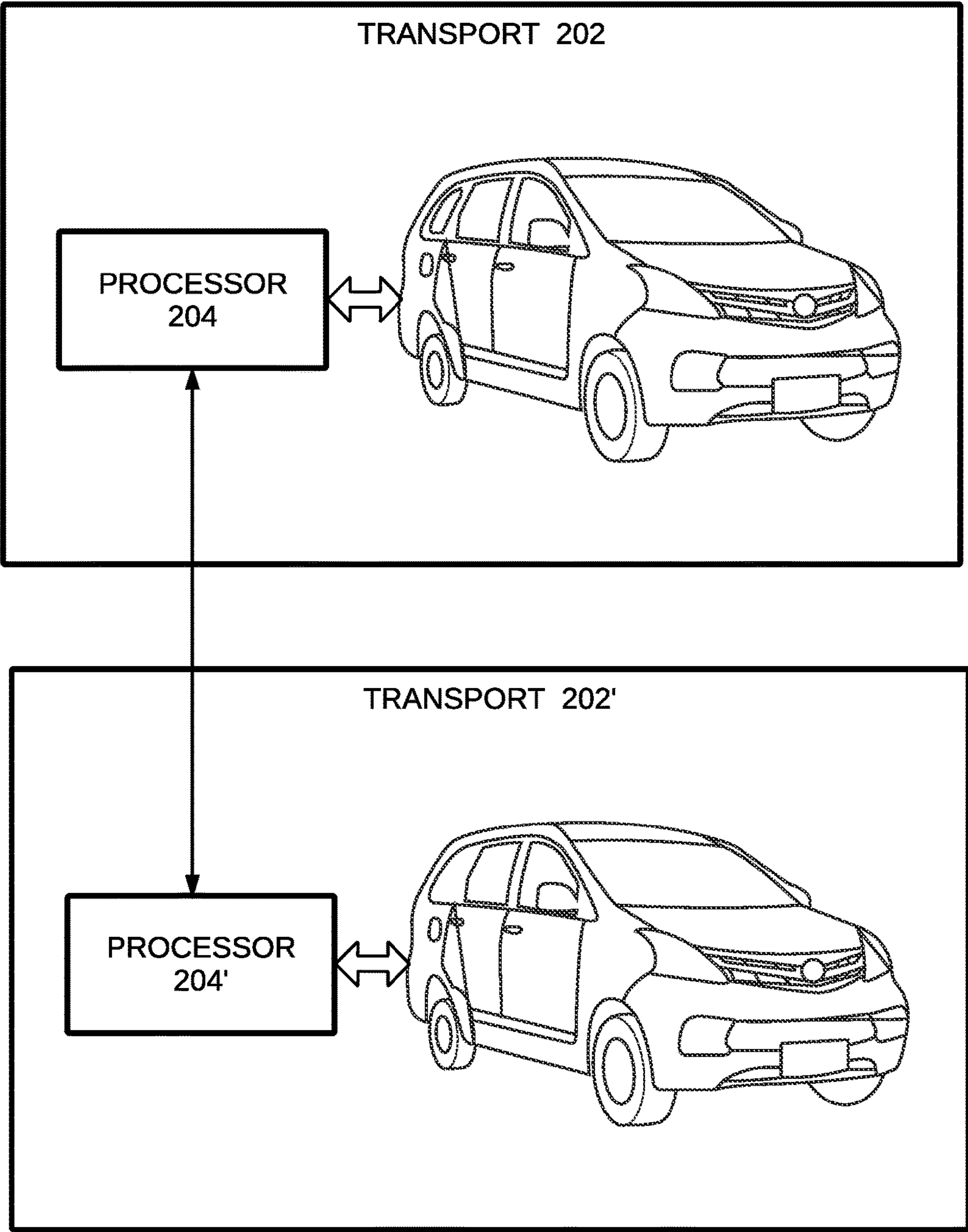


FIG. 2A

210

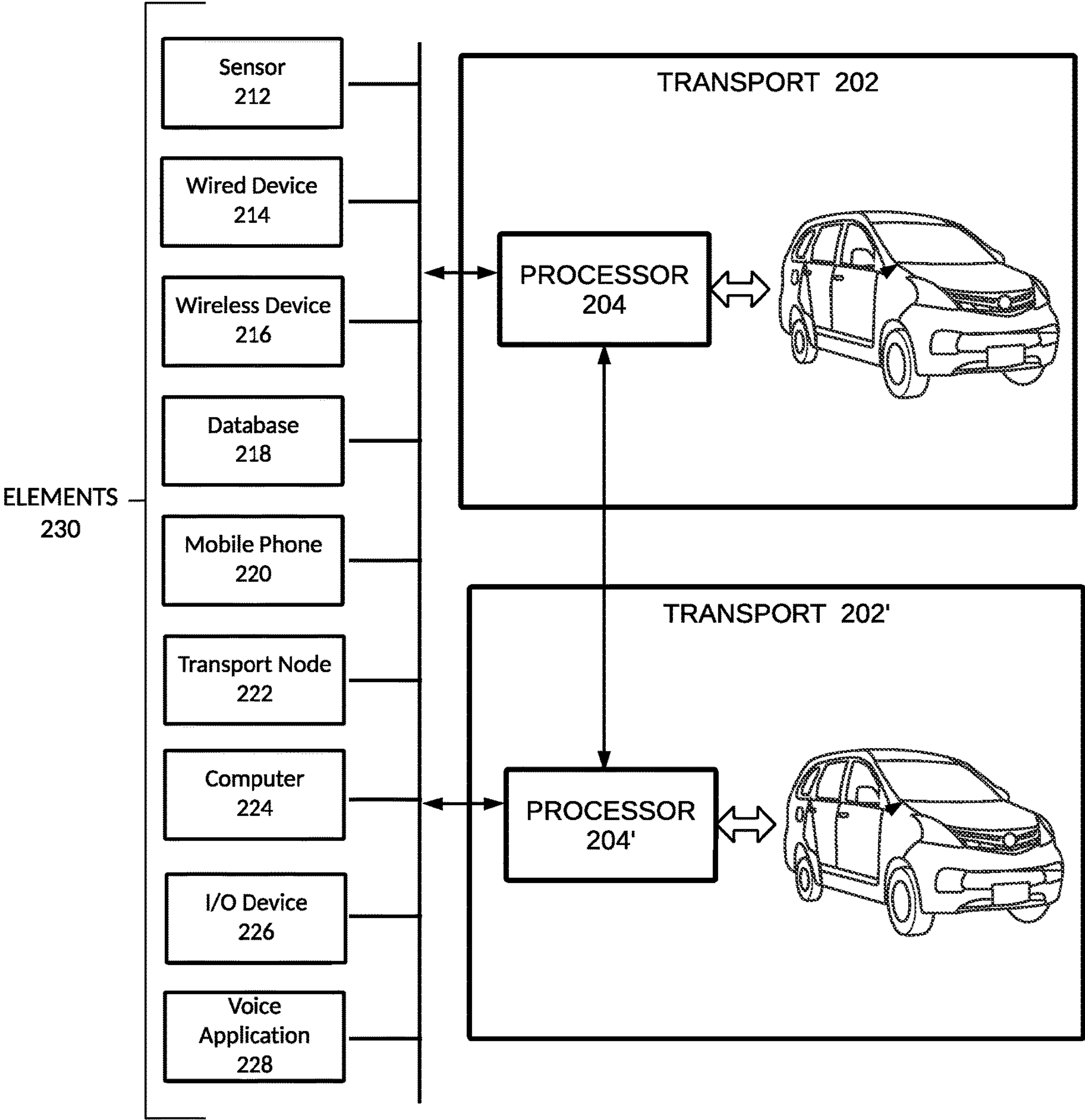


FIG. 2B

240

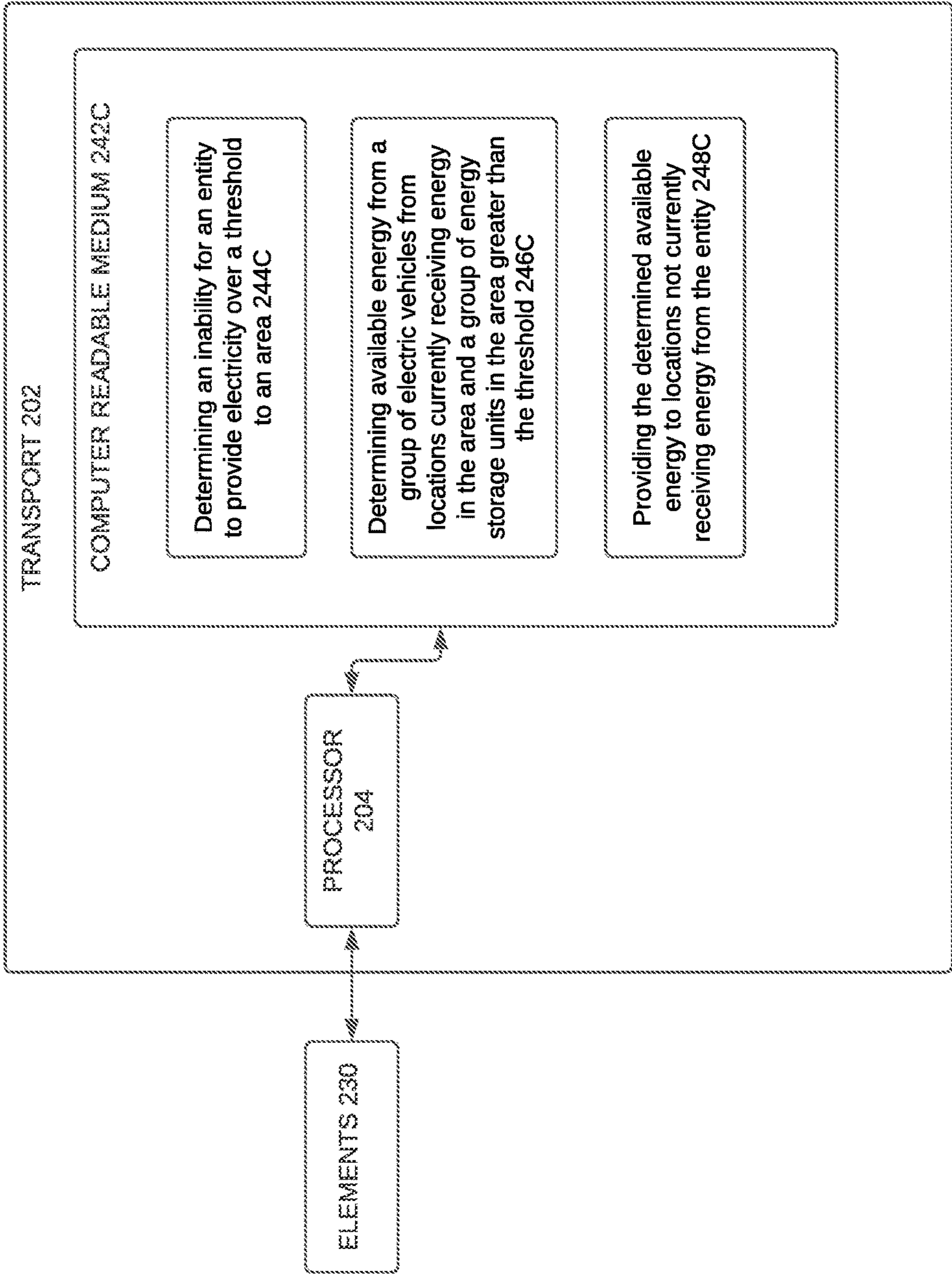


FIG. 2C

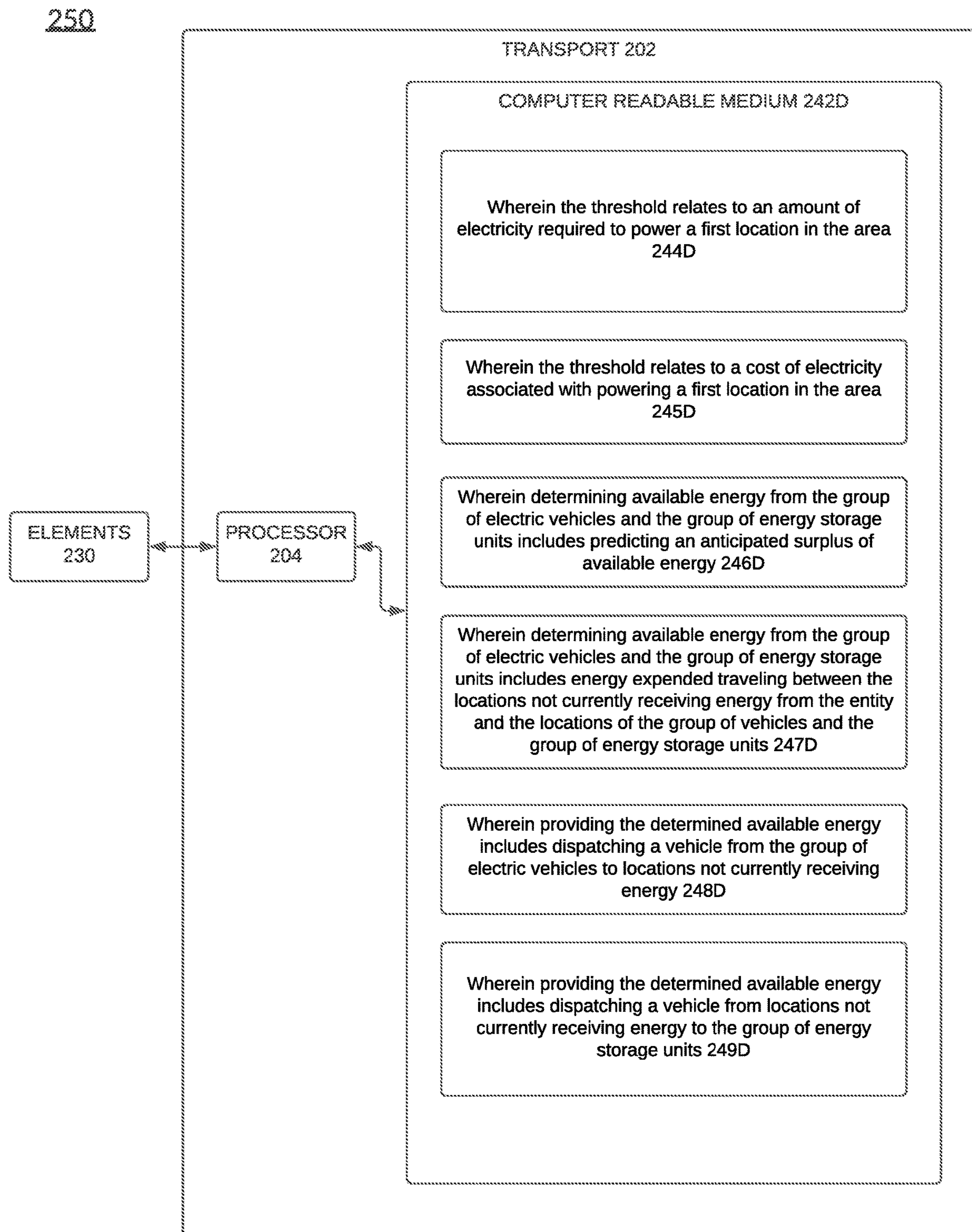


FIG. 2D

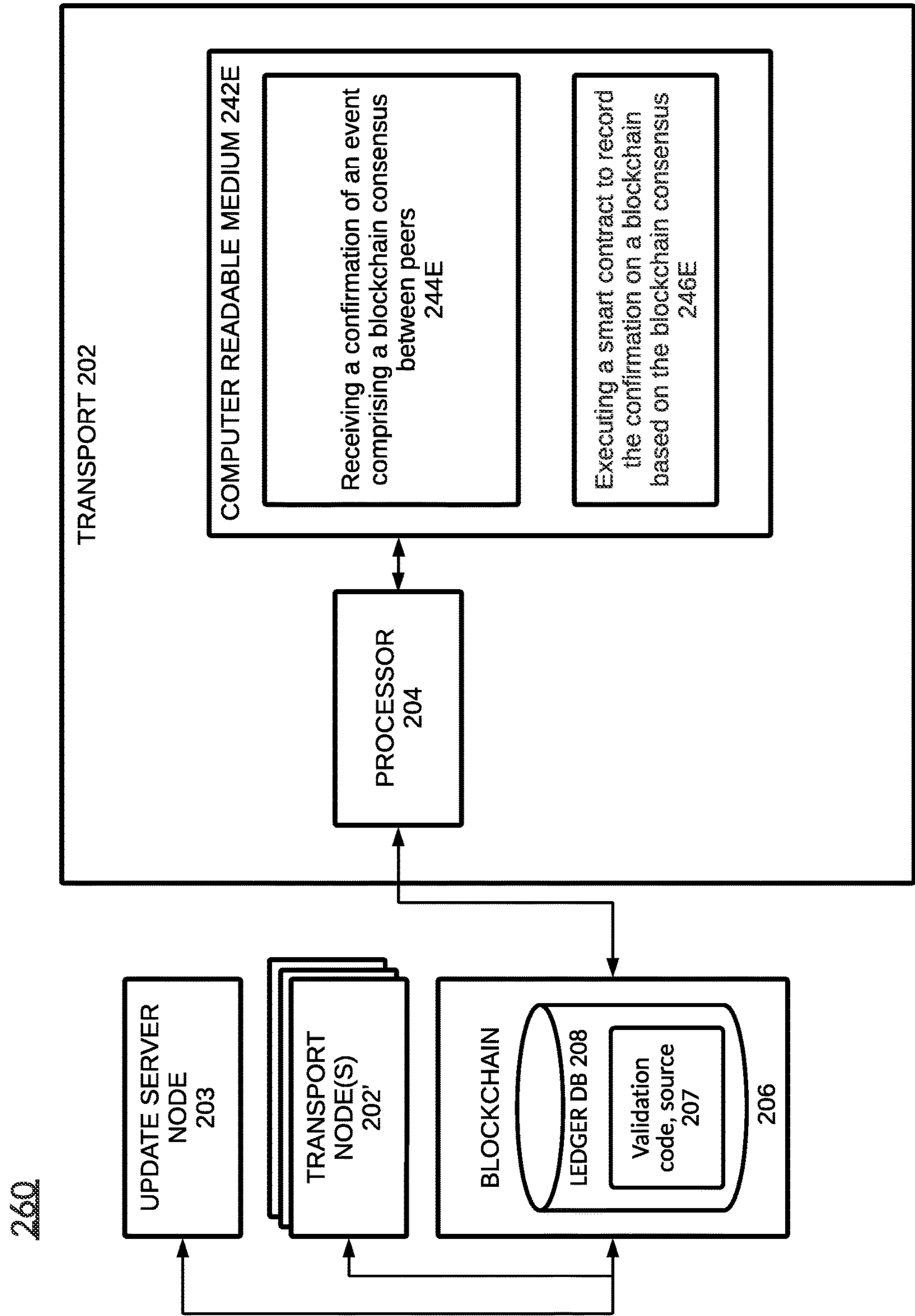


FIG. 2E

265

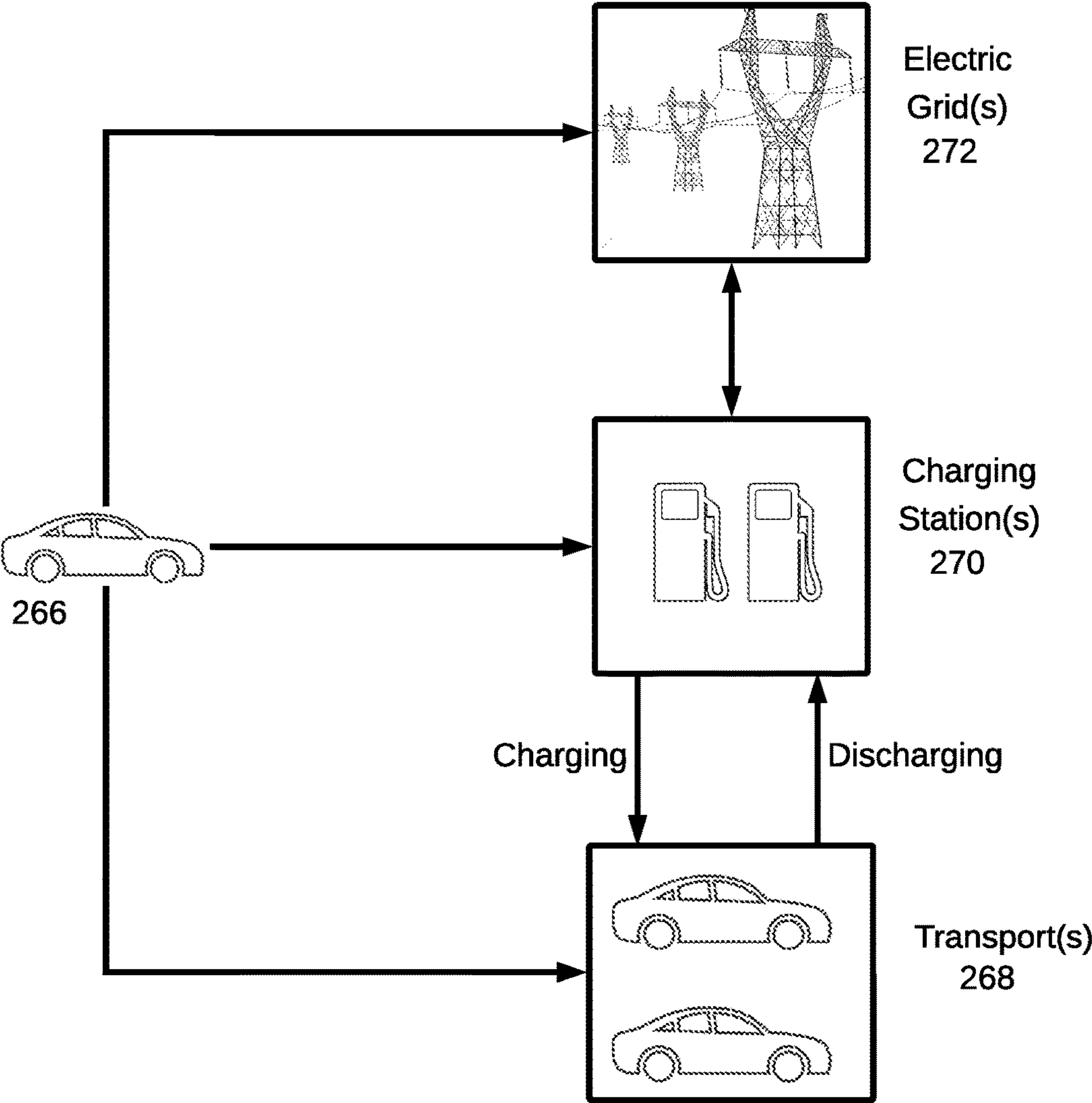


FIG. 2F

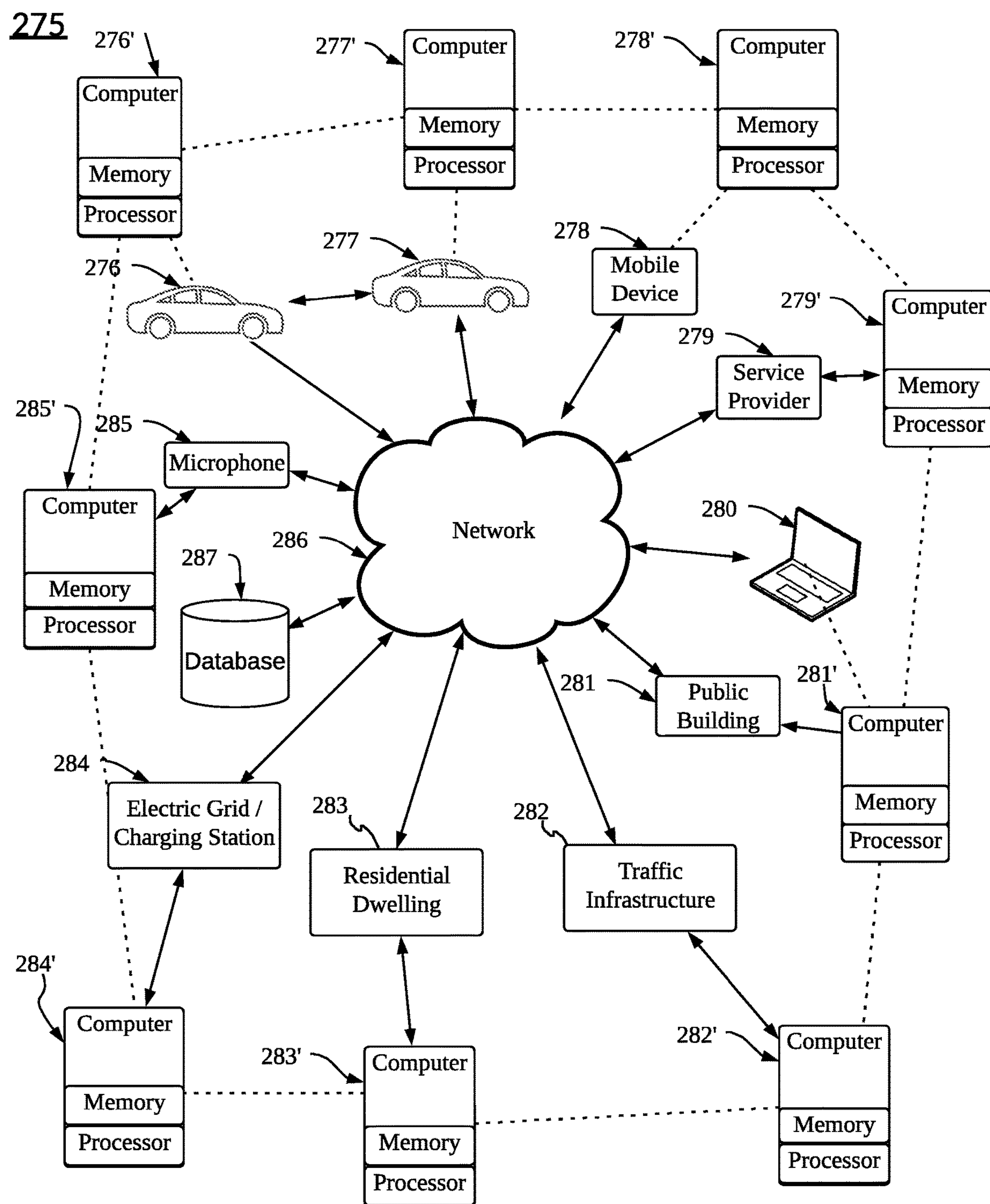


FIG. 2G

290

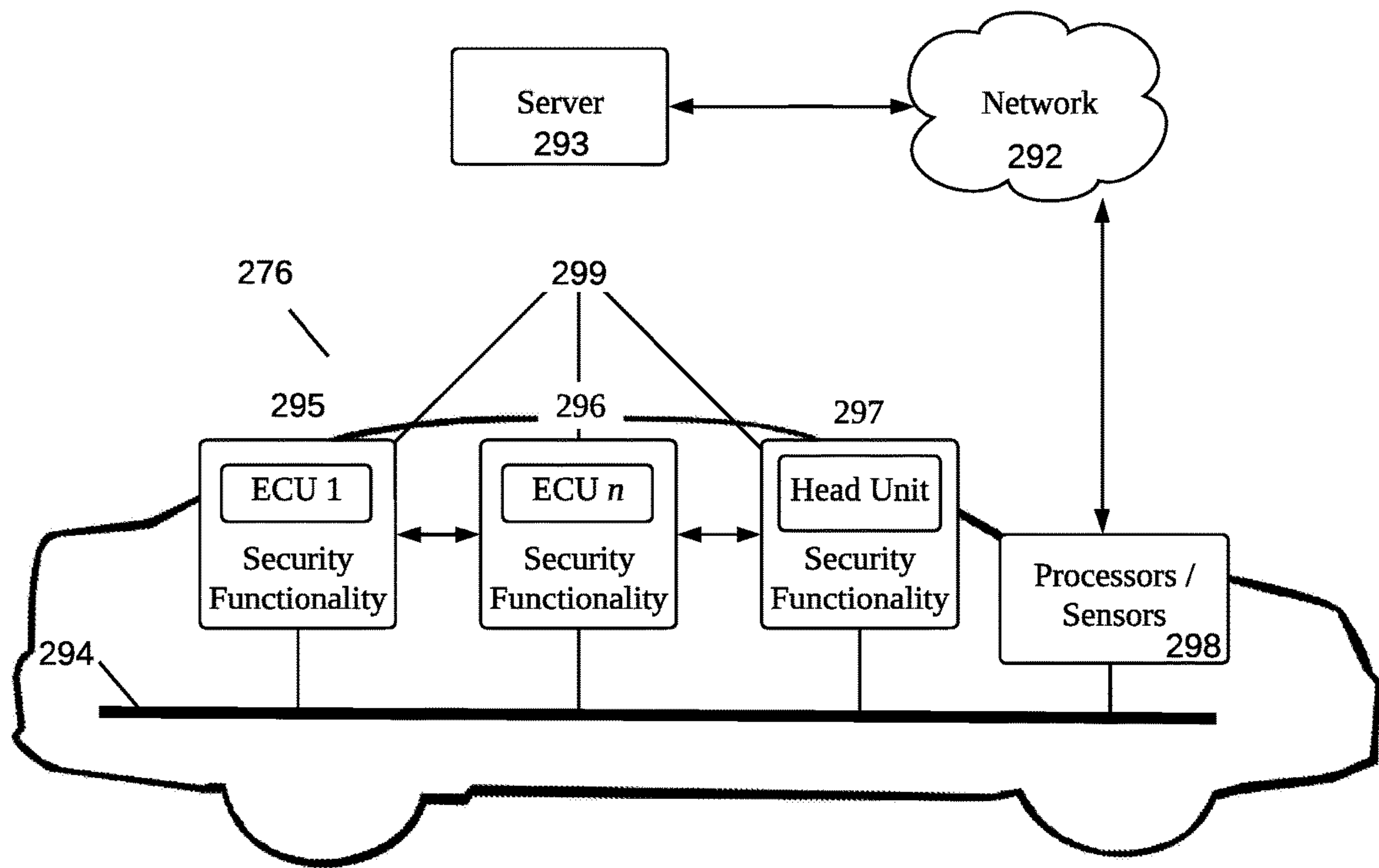


FIG. 2H

290A

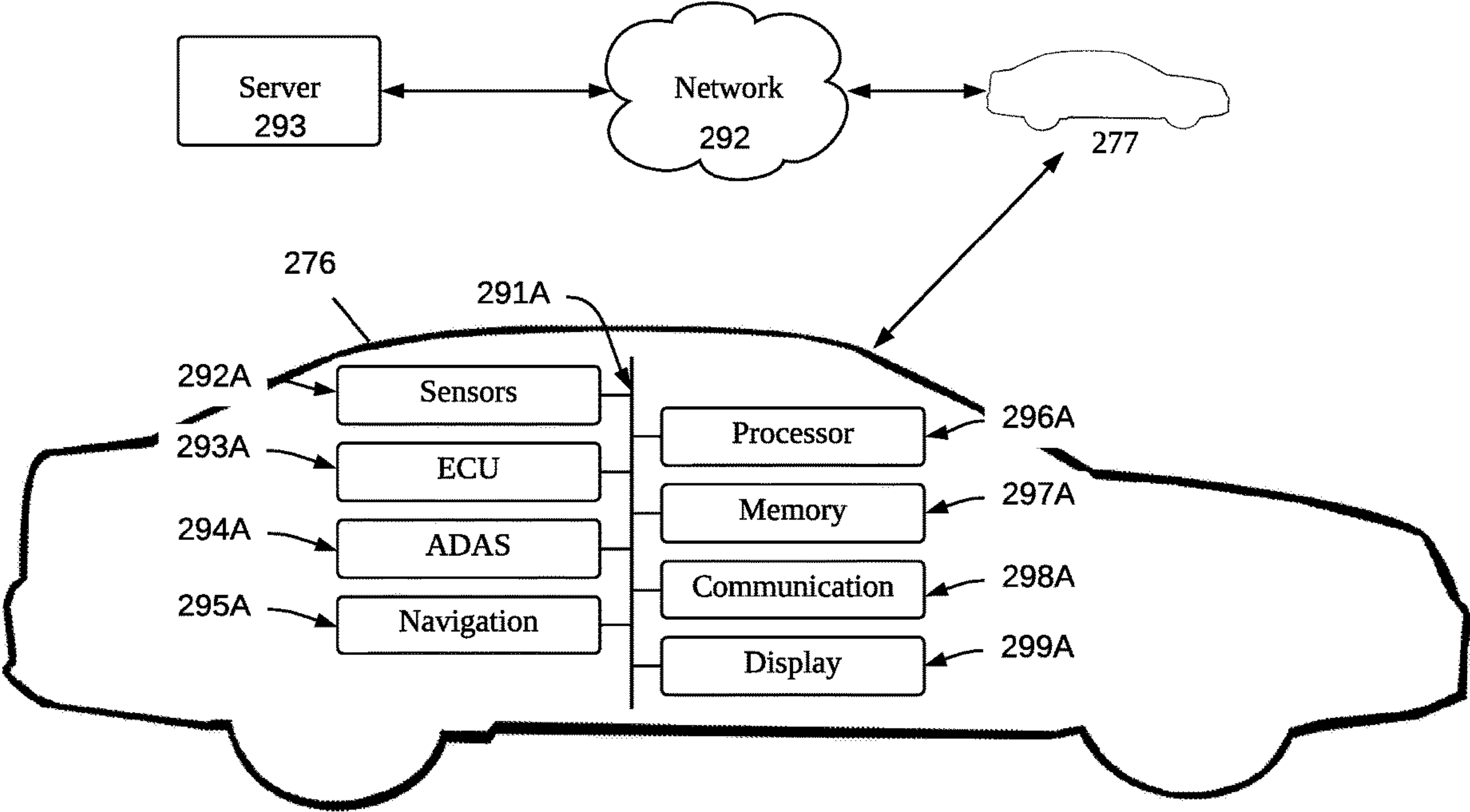


FIG. 2I

290B

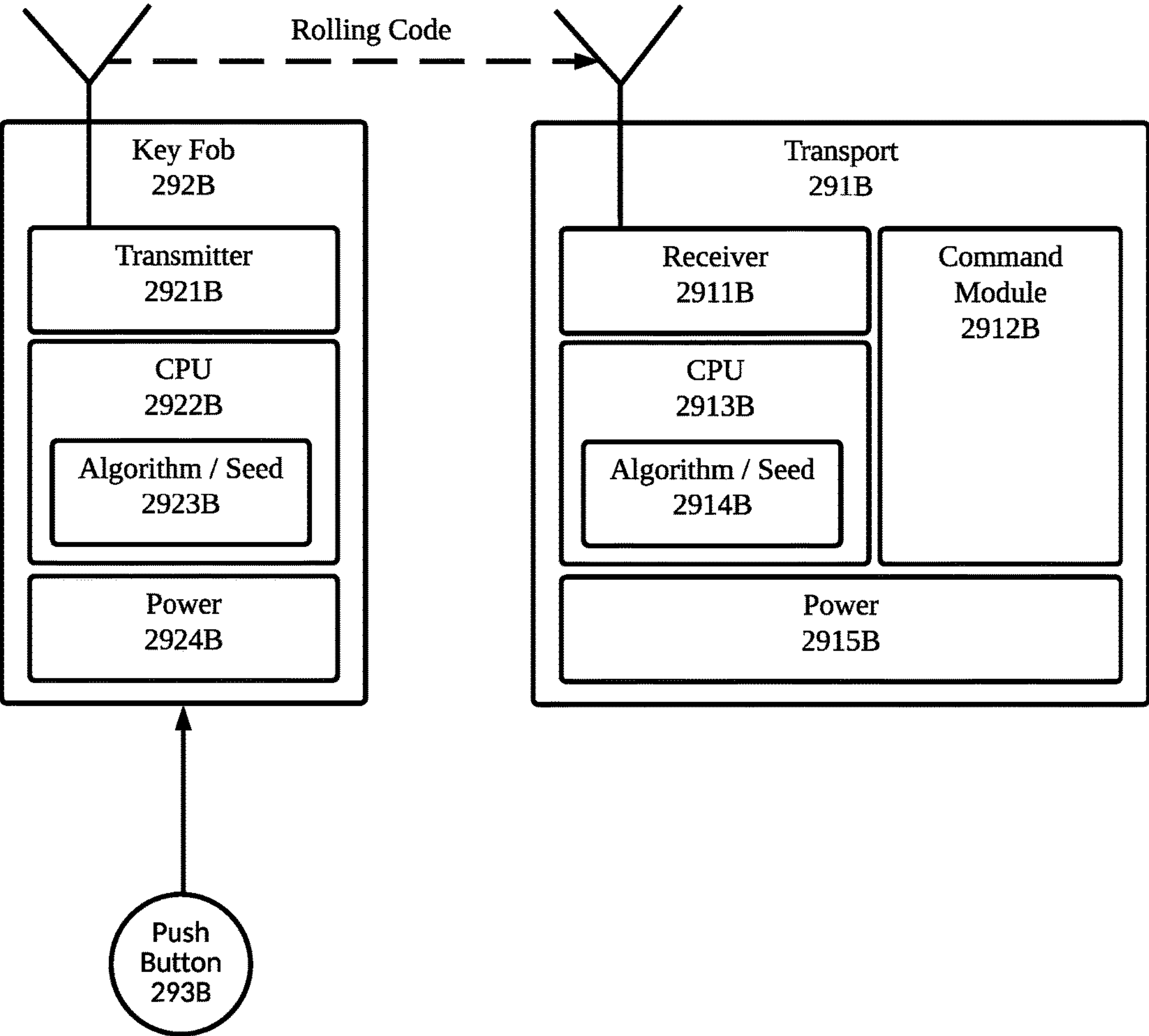


FIG. 2J

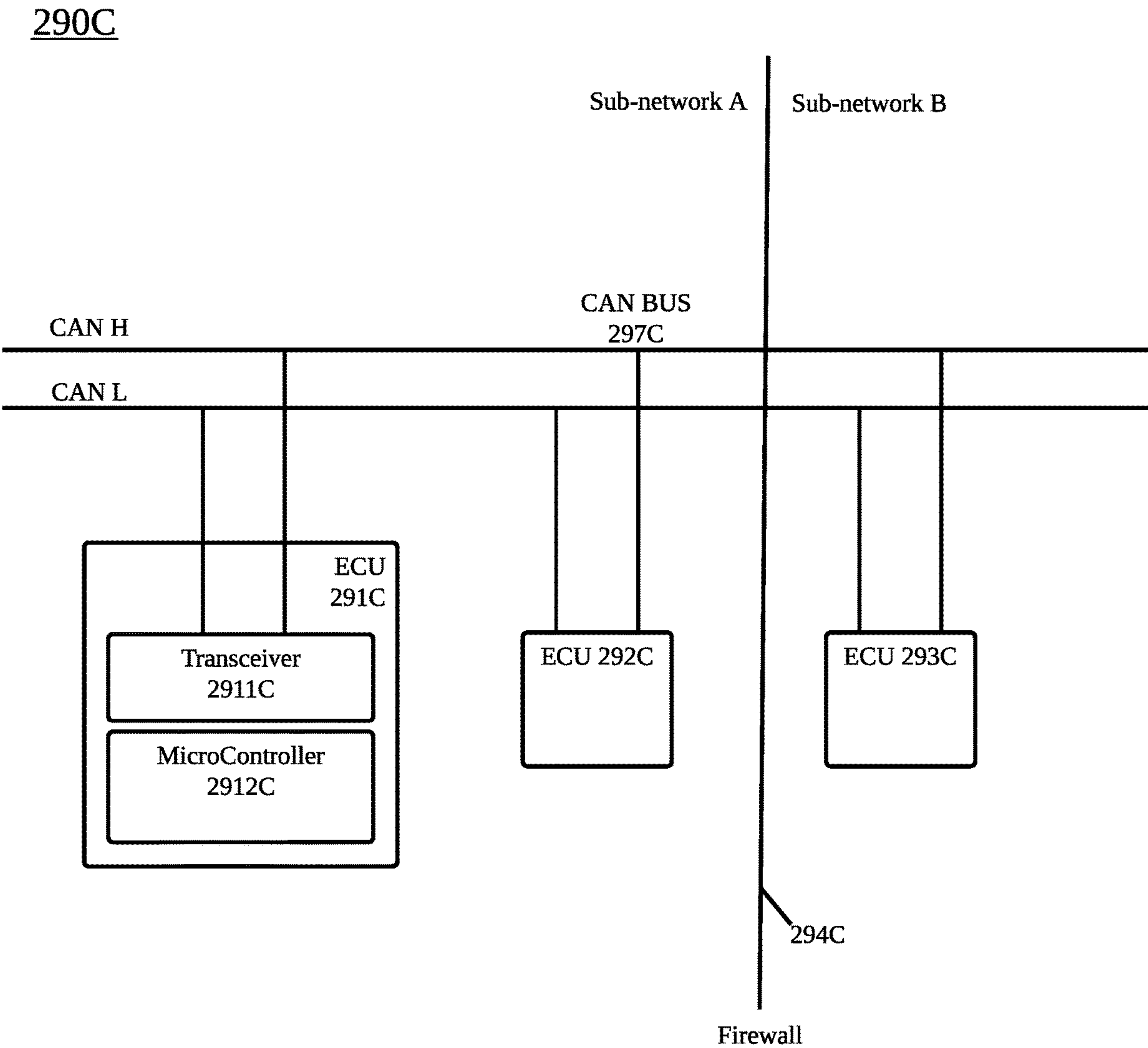


FIG. 2K

290D

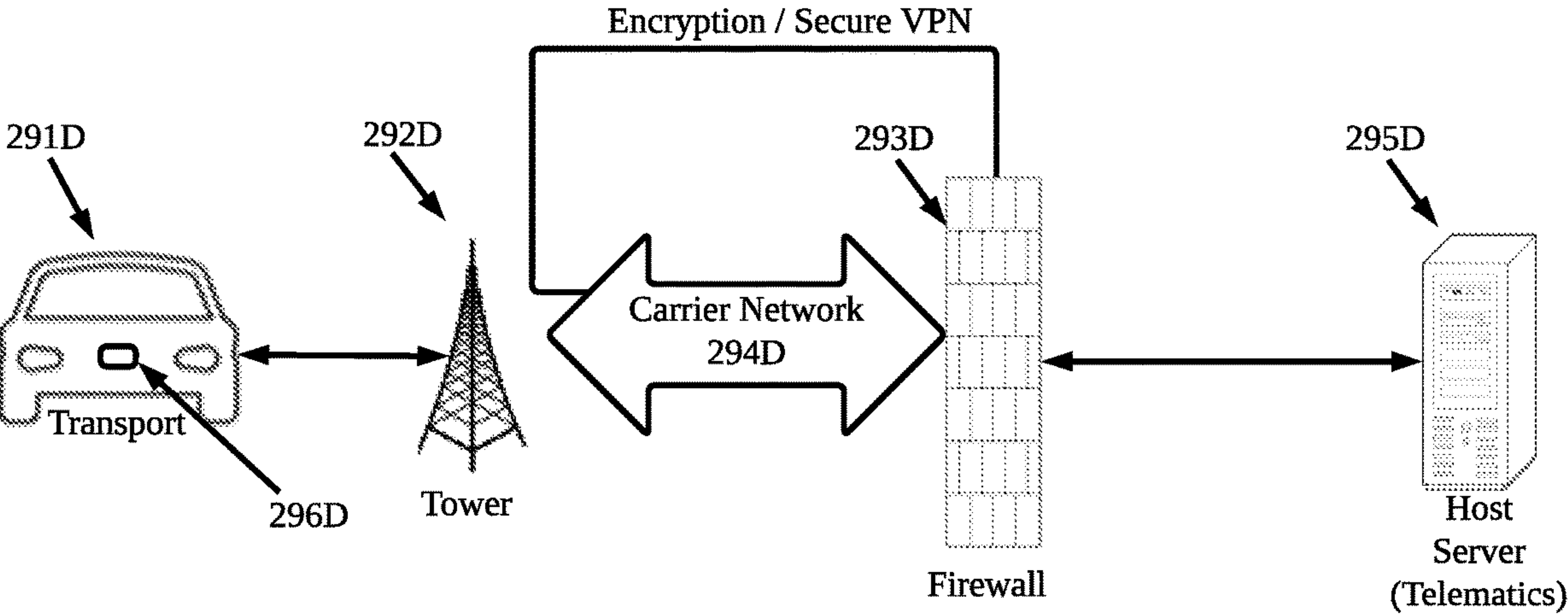


FIG. 2L

290E

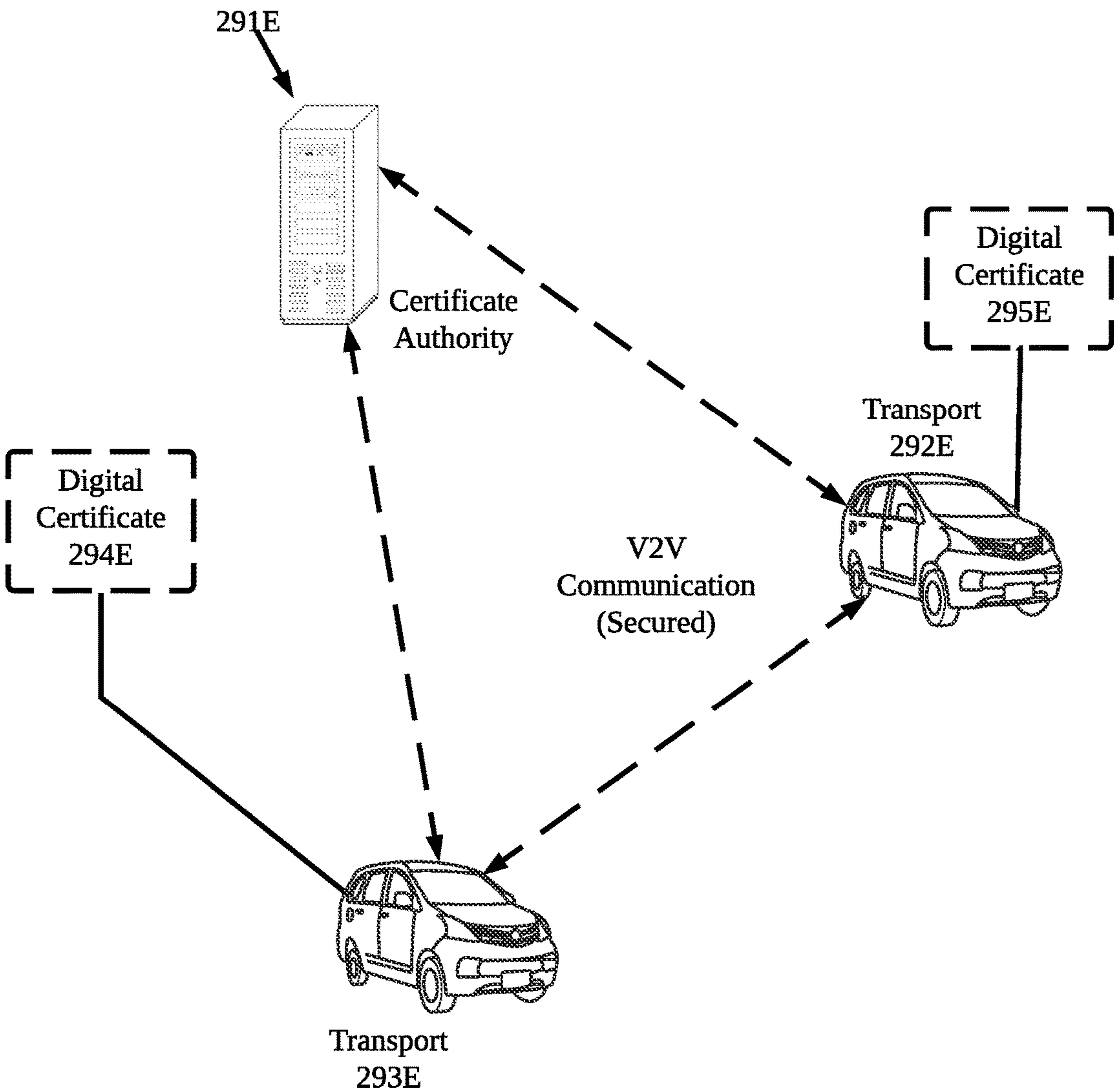


FIG. 2M

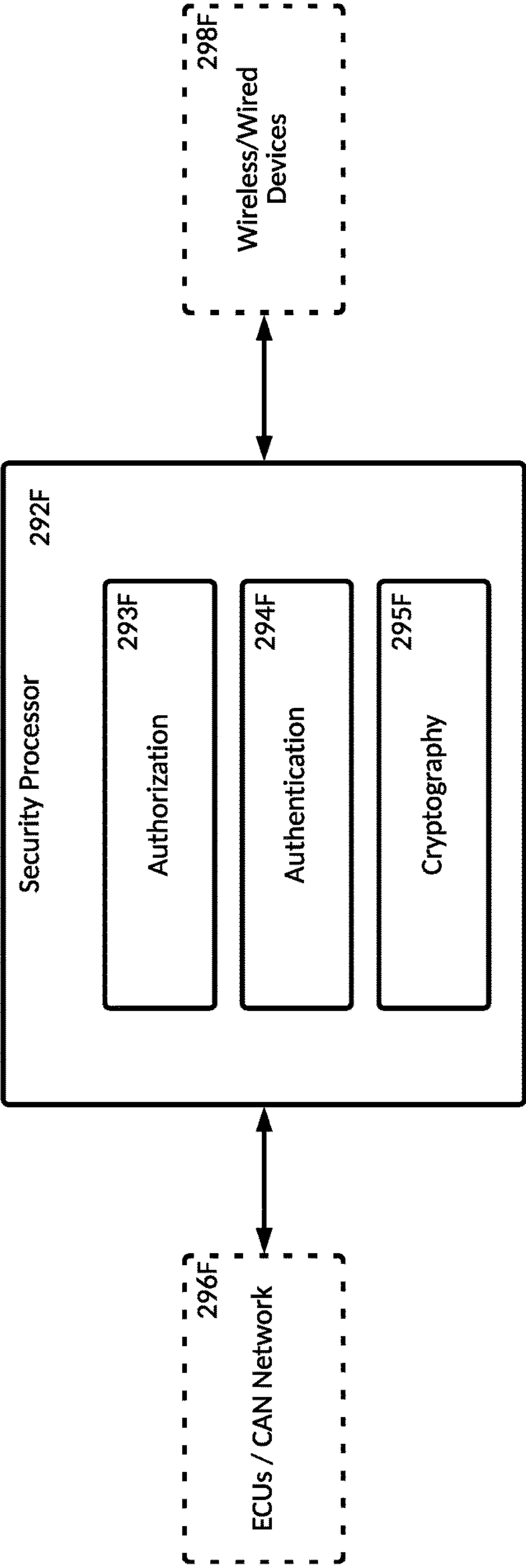


FIG. 2N

300

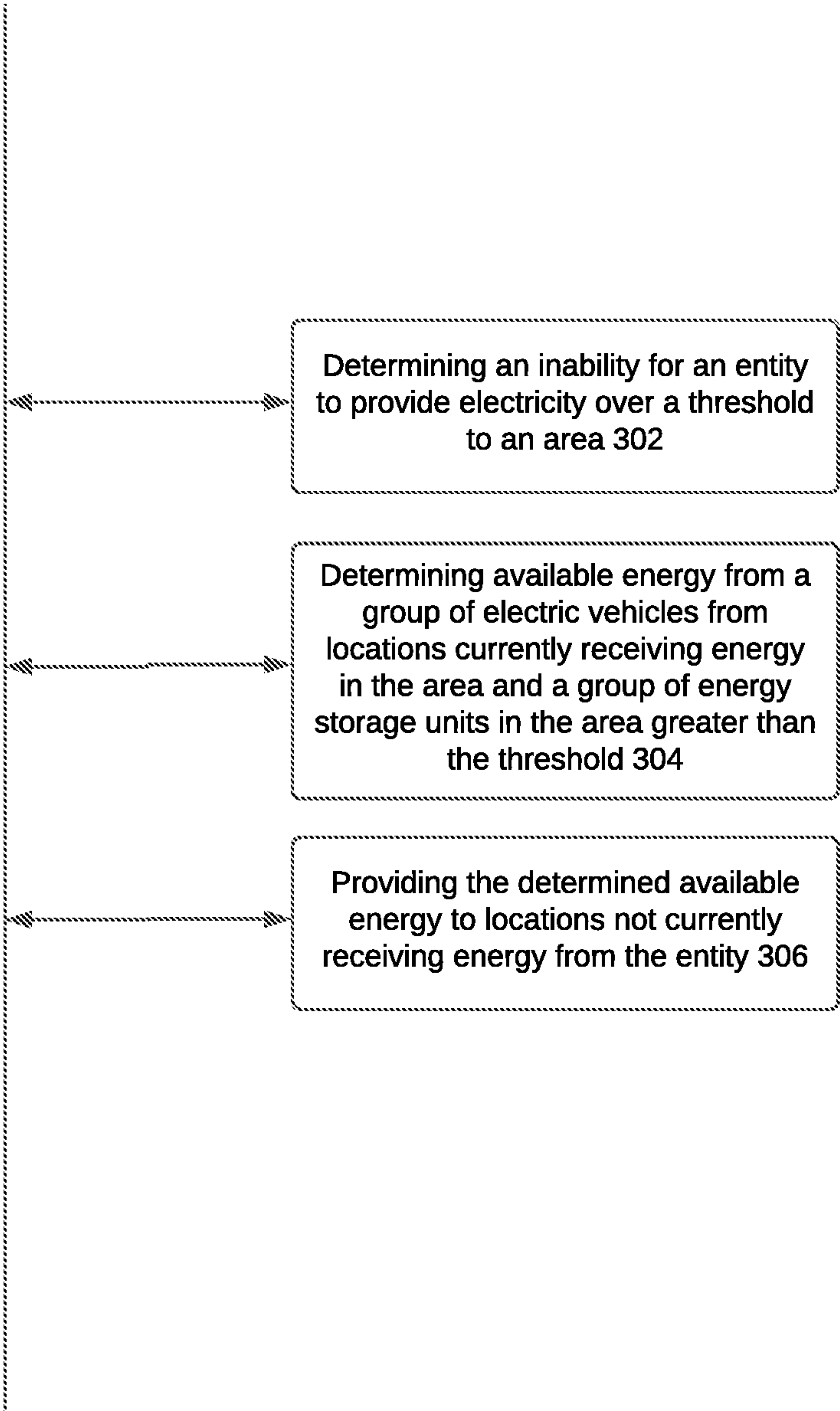


FIG. 3A

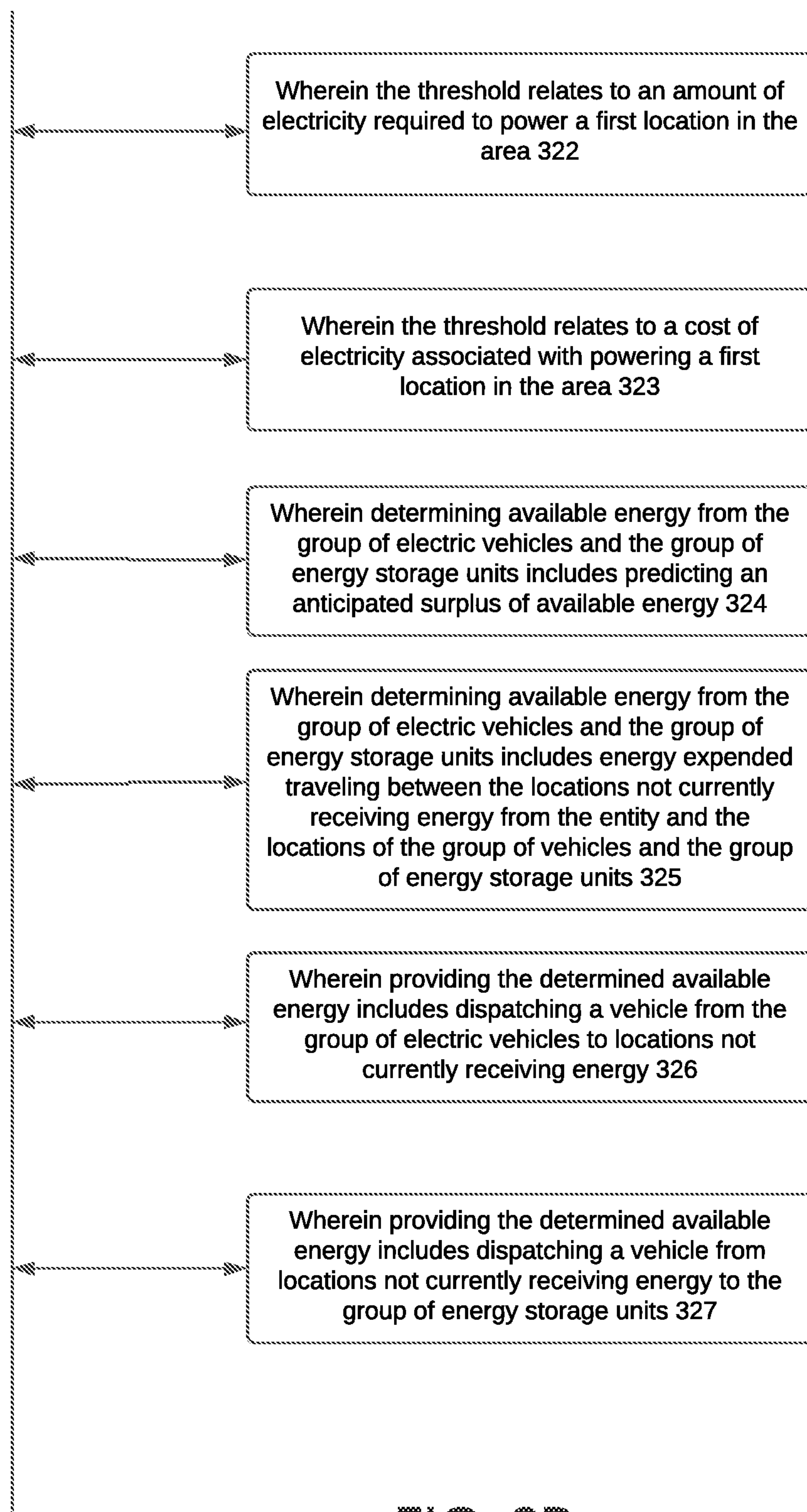
320

FIG. 3B

340

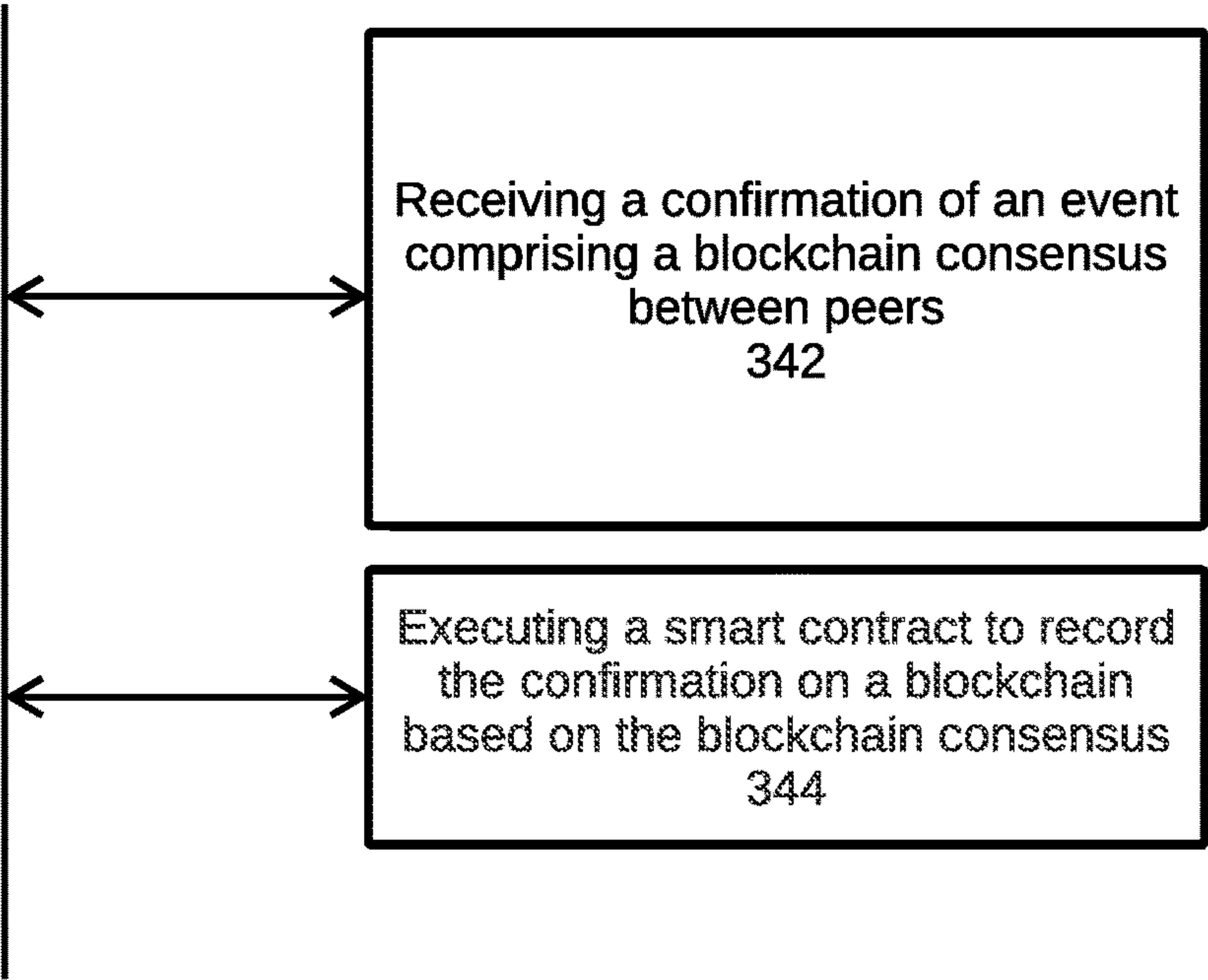


FIG. 3C

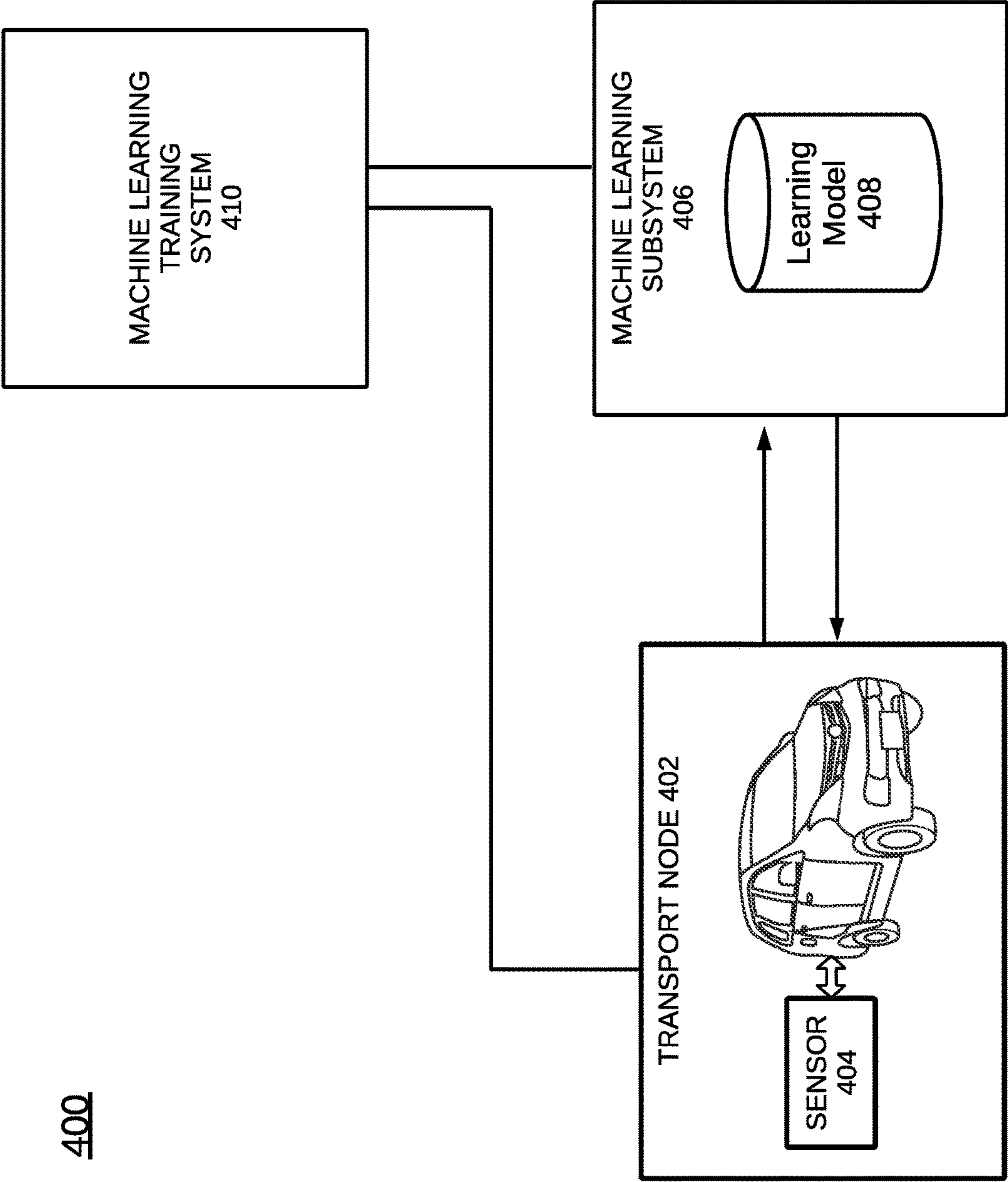


FIG. 4

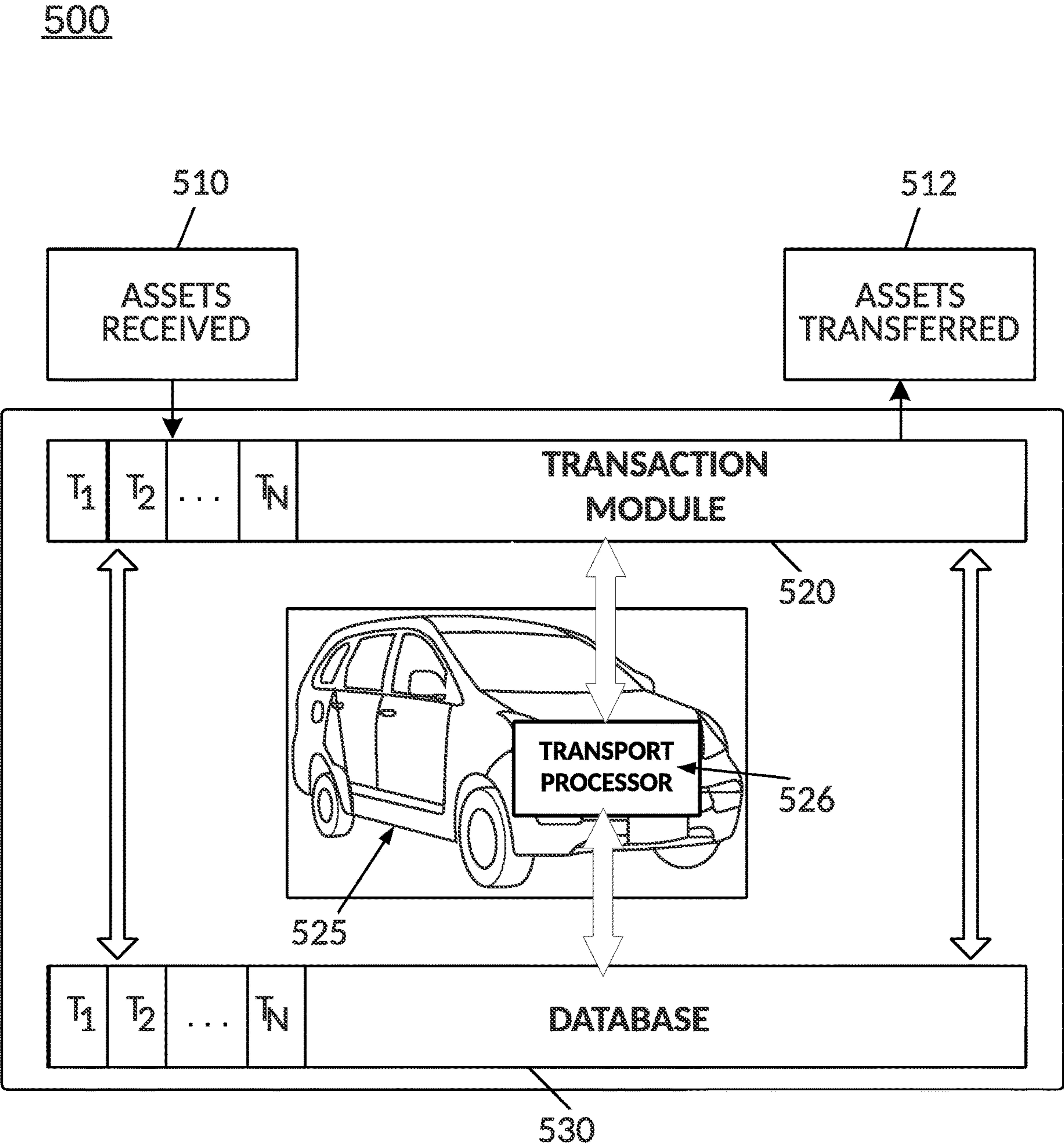


FIG. 5A

550

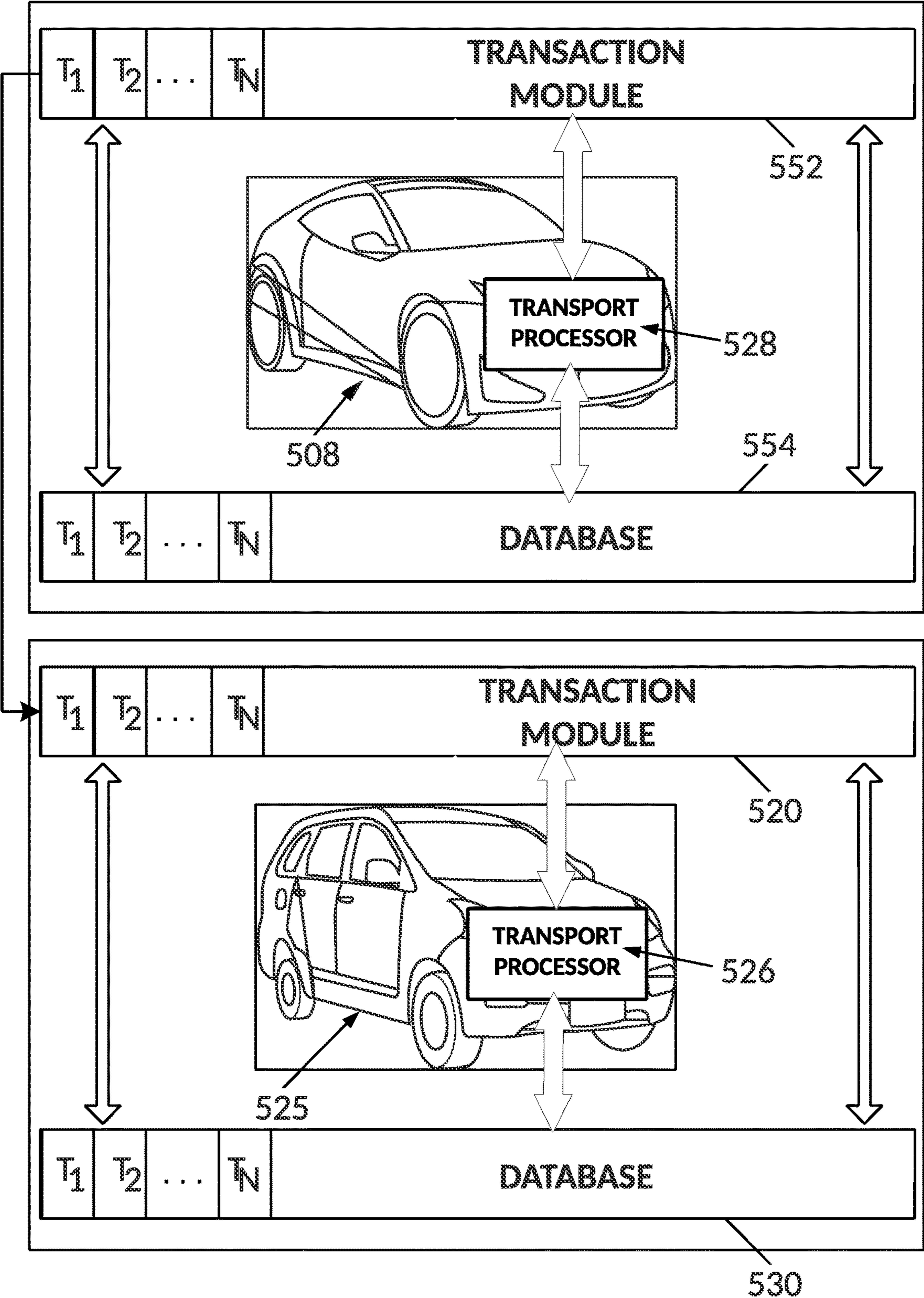


FIG. 5B

600

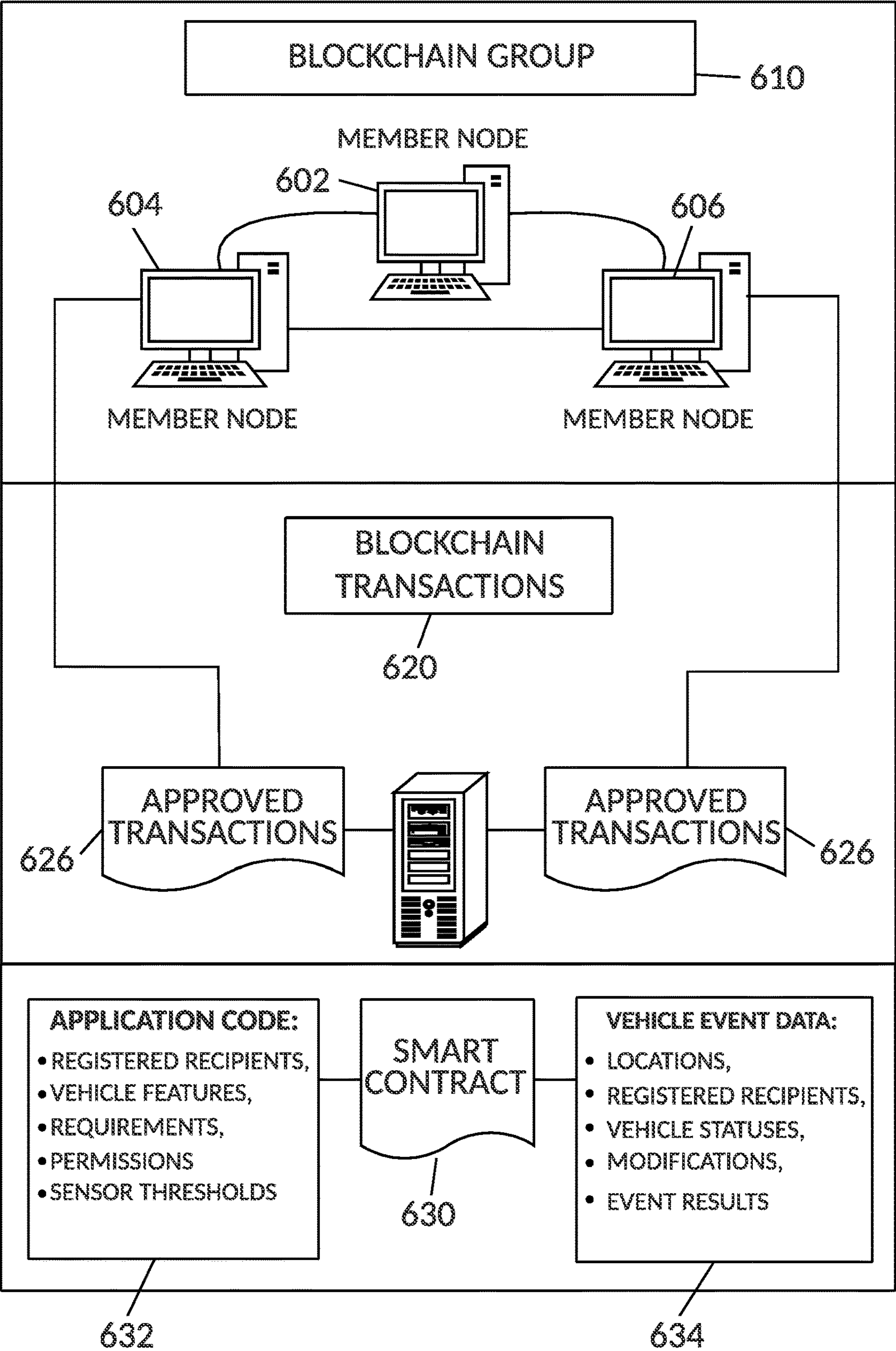


FIG. 6A

640

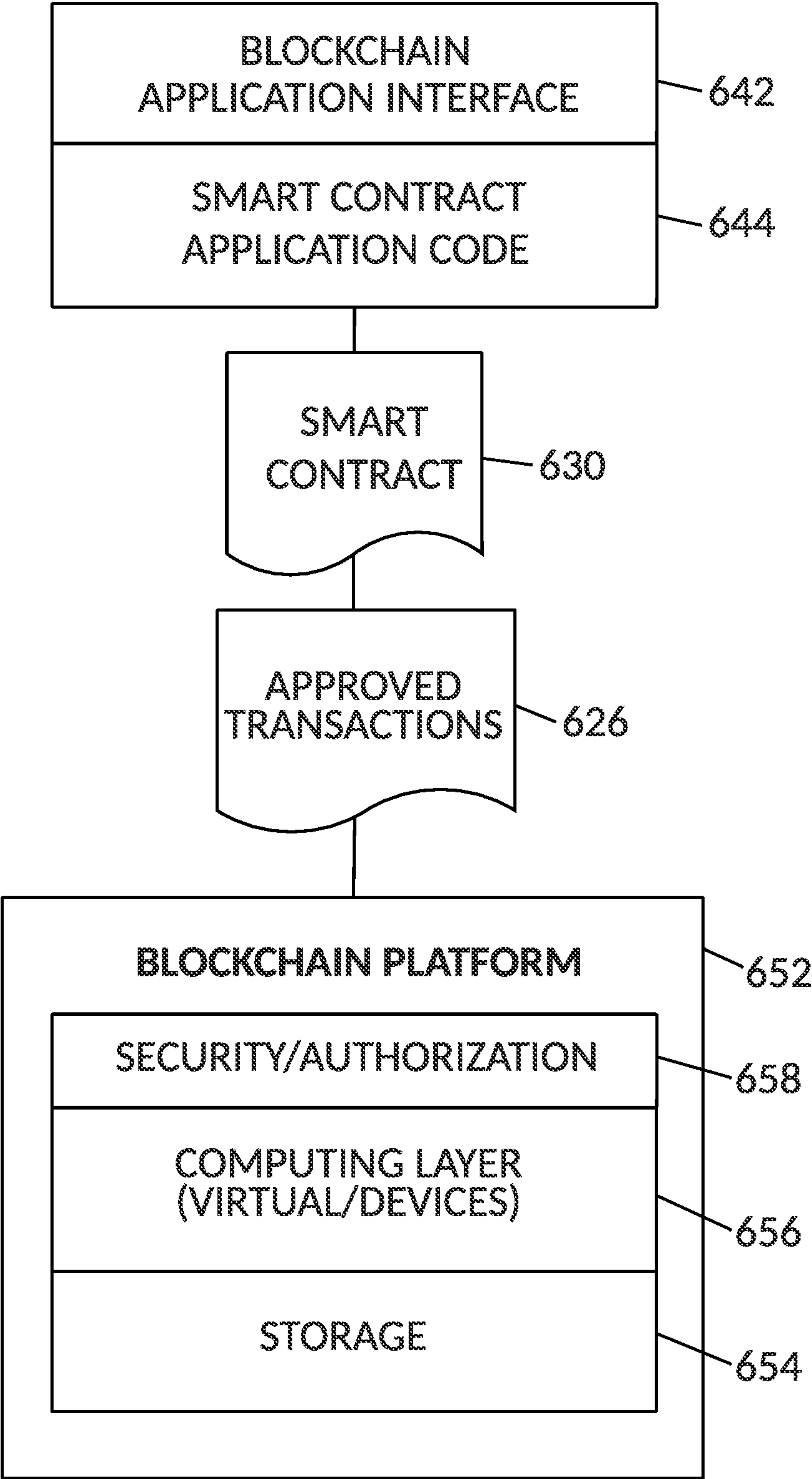


FIG. 6B

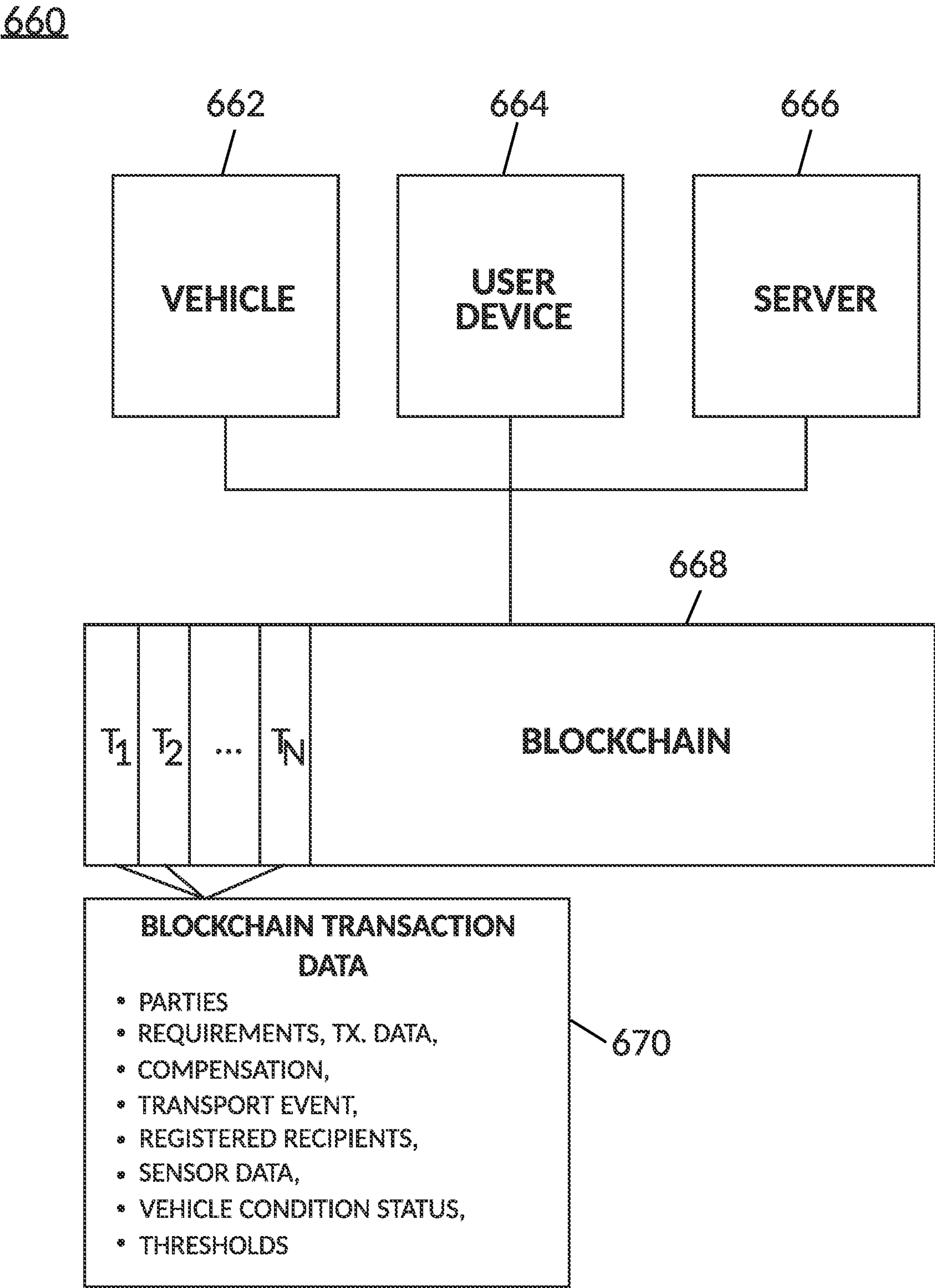
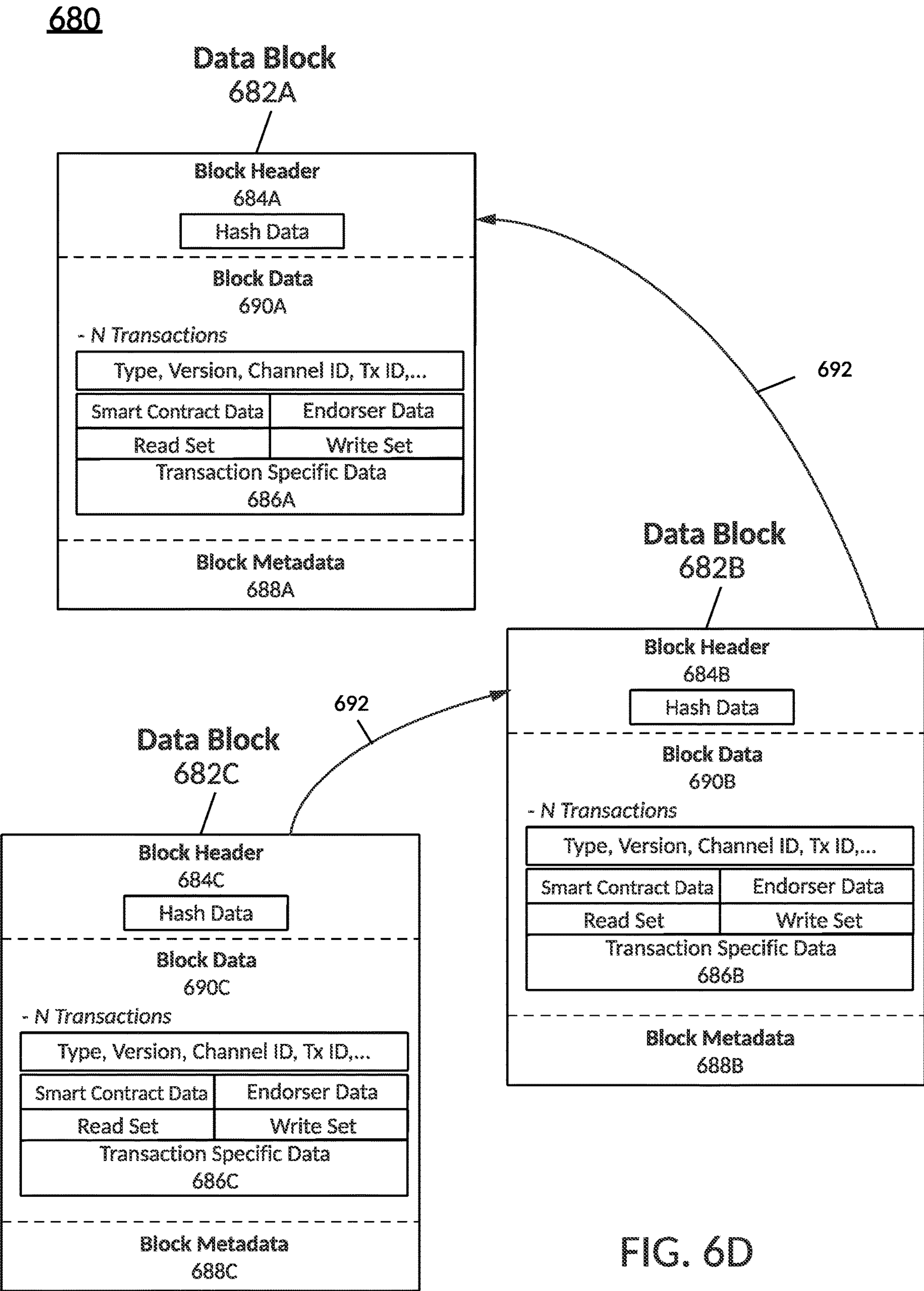


FIG. 6C



700

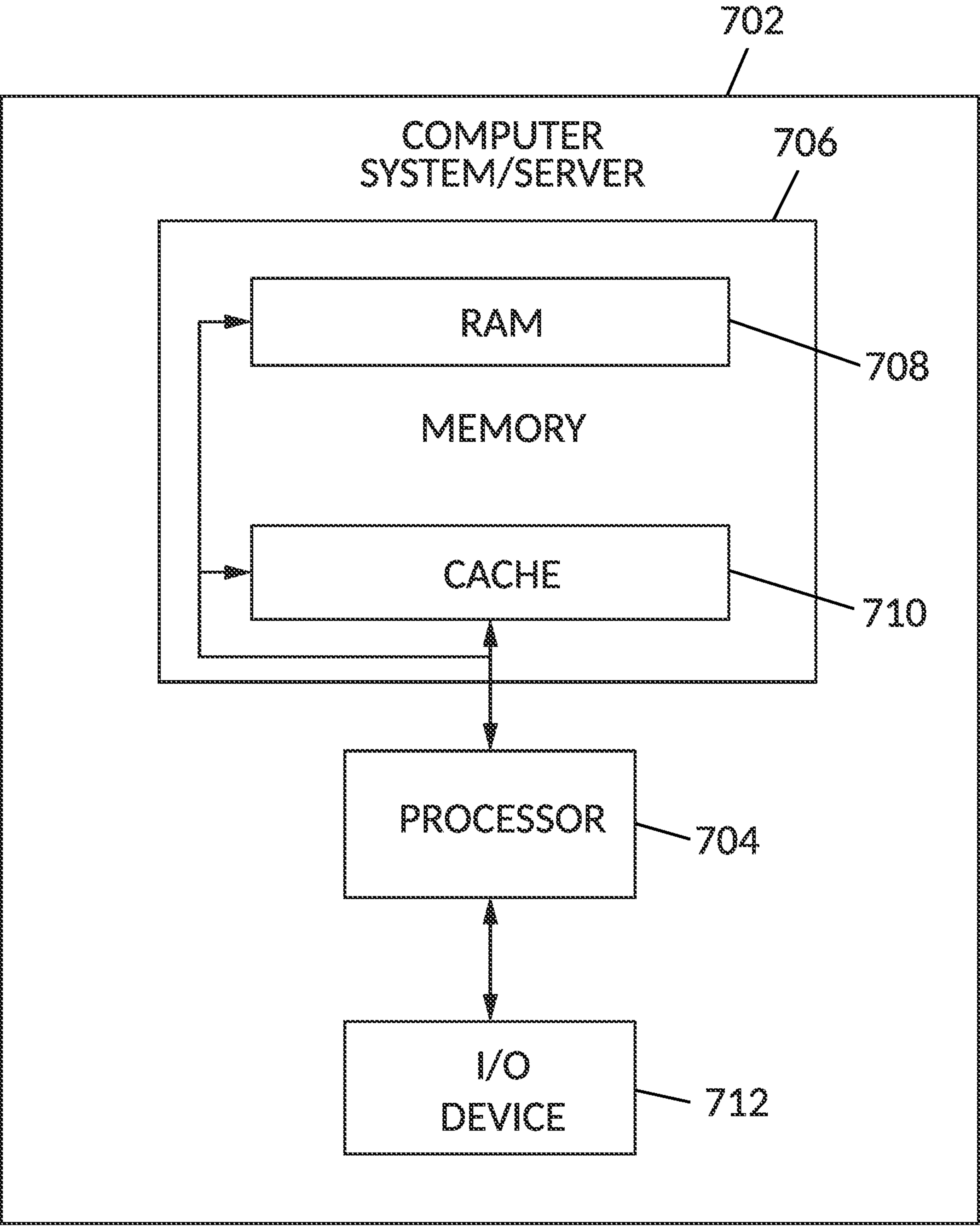


FIG. 7

DECENTRALIZED POWER EXCHANGE

BACKGROUND

[0001] Vehicles or transports, such as cars, motorcycles, trucks, planes, trains, etc., generally provide transportation needs to occupants and/or goods in a variety of ways. Functions related to transports may be identified and utilized by various computing devices, such as a smartphone or a computer located on and/or off the transport.

SUMMARY

[0002] One example embodiment provides a method that includes one or more of determining an inability for an entity to provide electricity over a threshold to an area; determining available energy from a group of electric vehicles from locations currently receiving energy in the area and a group of energy storage units in the area greater than the threshold; providing the determined available energy to locations not currently receiving energy from the entity.

[0003] Another example embodiment provides a system that includes a memory communicably coupled to a processor, wherein the processor performs one or more of determine an inability for an entity to provide electricity over a threshold to an area; determine available energy from a group of electric vehicles from locations currently receiving energy in the area and a group of energy storage units in the area greater than the threshold; provide the determined available energy to locations not currently receiving energy from the entity.

[0004] A further example embodiment provides a computer readable storage medium comprising instructions, that when read by a processor, cause the processor to perform one or more of determining an inability for an entity to provide electricity over a threshold to an area; determining available energy from a group of electric vehicles from locations currently receiving energy in the area and a group of energy storage units in the area greater than the threshold; providing the determined available energy to locations not currently receiving energy from the entity.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1A illustrates an example system to provide a decentralized power exchange, according to example embodiments.

[0006] FIG. 1B illustrates a further example system to provide a decentralized power exchange, according to example embodiments.

[0007] FIG. 1C illustrates a further example system to provide a decentralized power exchange, according to example embodiments.

[0008] FIG. 2A illustrates a transport network diagram, according to example embodiments.

[0009] FIG. 2B illustrates another transport network diagram, according to example embodiments.

[0010] FIG. 2C illustrates yet another transport network diagram, according to example embodiments.

[0011] FIG. 2D illustrates a further transport network diagram, according to example embodiments.

[0012] FIG. 2E illustrates yet a further transport network diagram, according to example embodiments.

[0013] FIG. 2F illustrates a diagram depicting electrification of one or more elements, according to example embodiments.

[0014] FIG. 2G illustrates a diagram depicting interconnections between different elements, according to example embodiments.

[0015] FIG. 2H illustrates a further diagram depicting interconnections between different elements, according to example embodiments.

[0016] FIG. 2I illustrates yet a further diagram depicting interconnections between elements, according to example embodiments.

[0017] FIG. 2J illustrates yet a further diagram depicting a keyless entry system, according to example embodiments.

[0018] FIG. 2K illustrates yet a further diagram depicting a CAN within a transport, according to example embodiments.

[0019] FIG. 2L illustrates yet a further diagram depicting an end-to-end communication channel, according to example embodiments.

[0020] FIG. 2M illustrates yet a further diagram depicting an example of transports performing secured V2V communications using security certificates, according to example embodiments.

[0021] FIG. 2N illustrates yet a further diagram depicting an example of a transport interacting with a security processor and a wireless device, according to example embodiments.

[0022] FIG. 3A illustrates a flow diagram, according to example embodiments.

[0023] FIG. 3B illustrates another flow diagram, according to example embodiments.

[0024] FIG. 3C illustrates yet another flow diagram, according to example embodiments.

[0025] FIG. 4 illustrates a machine learning transport network diagram, according to example embodiments.

[0026] FIG. 5A illustrates an example vehicle configuration for managing database transactions associated with a vehicle, according to example embodiments.

[0027] FIG. 5B illustrates another example vehicle configuration for managing database transactions conducted among various vehicles, according to example embodiments.

[0028] FIG. 6A illustrates a blockchain architecture configuration, according to example embodiments.

[0029] FIG. 6B illustrates another blockchain configuration, according to example embodiments.

[0030] FIG. 6C illustrates a blockchain configuration for storing blockchain transaction data, according to example embodiments.

[0031] FIG. 6D illustrates example data blocks, according to example embodiments.

[0032] FIG. 7 illustrates an example system that supports one or more of the example embodiments.

DETAILED DESCRIPTION

[0033] It will be readily understood that the instant components, as generally described and illustrated in the figures herein, may be arranged and designed in a wide variety of different configurations. Thus, the following detailed description of the embodiments of at least one of a method, apparatus, computer readable storage medium and system, as represented in the attached figures, is not intended to limit the scope of the application as claimed but is merely representative of selected embodiments. Multiple embodiments depicted herein are not intended to limit the scope of the solution. The computer-readable storage medium may be

a non-transitory computer readable medium or a non-transitory computer readable storage medium.

[0034] Communications between the transport(s) and certain entities, such as remote servers, other transports and local computing devices (e.g., smartphones, personal computers, transport-embedded computers, etc.) may be sent and/or received and processed by one or more ‘components’ which may be hardware, firmware, software or a combination thereof. The components may be part of any of these entities or computing devices or certain other computing devices. In one example, consensus decisions related to blockchain transactions may be performed by one or more computing devices or components (which may be any element described and/or depicted herein) associated with the transport(s) and one or more of the components outside or at a remote location from the transport(s).

[0035] The instant features, structures, or characteristics described in this specification may be combined in any suitable manner in one or more embodiments. For example, the usage of the phrases “example embodiments,” “some embodiments,” or other similar language, throughout this specification refers to the fact that a particular feature, structure, or characteristic described in connection with the embodiment may be included in at least one example. Thus, appearances of the phrases “example embodiments”, “in some embodiments”, “in other embodiments,” or other similar language, throughout this specification do not necessarily all refer to the same group of embodiments, and the described features, structures, or characteristics may be combined in any suitable manner in one or more embodiments. In the diagrams, any connection between elements can permit one-way and/or two-way communication, even if the depicted connection is a one-way or two-way arrow. In the current solution, a vehicle or transport may include one or more of cars, trucks, walking area battery electric vehicle (BEV), e-Palette, fuel cell bus, motorcycles, scooters, bicycles, boats, recreational vehicles, planes, and any object that may be used to transport people and or goods from one location to another.

[0036] In addition, while the term “message” may have been used in the description of embodiments, other types of network data, such as, a packet, frame, datagram, etc. may also be used. Furthermore, while certain types of messages and signaling may be depicted in exemplary embodiments they are not limited to a certain type of message and signaling.

[0037] Example embodiments provide methods, systems, components, non-transitory computer readable medium, devices, and/or networks, which provide at least one of a transport (also referred to as a vehicle or car herein), a data collection system, a data monitoring system, a verification system, an authorization system, and a vehicle data distribution system. The vehicle status condition data received in the form of communication messages, such as wireless data network communications and/or wired communication messages, may be processed to identify vehicle/transport status conditions and provide feedback on the condition and/or changes of a transport. In one example, a user profile may be applied to a particular transport/vehicle to authorize a current vehicle event, service stops at service stations, to authorize subsequent vehicle rental services, and enable vehicle-to-vehicle communications.

[0038] Within the communication infrastructure, a decentralized database is a distributed storage system which

includes multiple nodes that communicate with each other. A blockchain is an example of a decentralized database, which includes an append-only immutable data structure (i.e., a distributed ledger) capable of maintaining records between untrusted parties. The untrusted parties are referred to herein as peers, nodes, or peer nodes. Each peer maintains a copy of the database records, and no single peer can modify the database records without a consensus being reached among the distributed peers. For example, the peers may execute a consensus protocol to validate blockchain storage entries, group the storage entries into blocks, and build a hash chain via the blocks. This process forms the ledger by ordering the storage entries, as is necessary, for consistency. In public or permissionless blockchains, anyone can participate without a specific identity. Public blockchains can involve crypto-currencies and use consensus-based on various protocols such as proof of work (PoW). Conversely, a permissioned blockchain database can secure interactions among a group of entities, which share a common goal, but which do not or cannot fully trust one another, such as businesses that exchange funds, goods, information, and the like. The instant solution can function in a permissioned and/or a permissionless blockchain setting.

[0039] Smart contracts are trusted distributed applications which leverage tamper-proof properties of the shared or distributed ledger (which may be in the form of a blockchain) and an underlying agreement between member nodes, which is referred to as an endorsement or endorsement policy. In general, blockchain entries are “endorsed” before being committed to the blockchain while entries, which are not endorsed are disregarded. A typical endorsement policy allows smart contract executable code to specify endorsers for an entry in the form of a set of peer nodes that are necessary for endorsement. When a client sends the entry to the peers specified in the endorsement policy, the entry is executed to validate the entry. After validation, the entries enter an ordering phase in which a consensus protocol produces an ordered sequence of endorsed entries grouped into blocks.

[0040] Nodes are the communication entities of the blockchain system. A “node” may perform a logical function in the sense that multiple nodes of different types can run on the same physical server. Nodes are grouped in trust domains and are associated with logical entities that control them in various ways. Nodes may include different types, such as a client or submitting-client node, which submits an entry-invocation to an endorser (e.g., peer), and broadcasts entry proposals to an ordering service (e.g., ordering node). Another type of node is a peer node, which can receive client submitted entries, commit the entries and maintain a state and a copy of the ledger of blockchain entries. Peers can also have the role of an endorser. An ordering-service-node or orderer is a node running the communication service for all nodes and which implements a delivery guarantee, such as a broadcast to each of the peer nodes in the system when committing entries and modifying a world state of the blockchain. The world state can constitute the initial blockchain entry, which normally includes control and setup information.

[0041] A ledger is a sequenced, tamper-resistant record of all state transitions of a blockchain. State transitions may result from smart contract executable code invocations (i.e., entries) submitted by participating parties (e.g., client nodes, ordering nodes, endorser nodes, peer nodes, etc.). An entry

may result in a set of asset key-value pairs being committed to the ledger as one or more operands, such as creates, updates, deletes, and the like. The ledger includes a blockchain (also referred to as a chain), which stores an immutable, sequenced record in blocks. The ledger also includes a state database, which maintains a current state of the blockchain. There is typically one ledger per channel. Each peer node maintains a copy of the ledger for each channel of which they are a member.

[0042] A chain is an entry log structured as hash-linked blocks, and each block contains a sequence of N entries where N is equal to or greater than one. The block header includes a hash of the blocks' entries, as well as a hash of the prior block's header. In this way, all entries on the ledger may be sequenced and cryptographically linked together. Accordingly, it is not possible to tamper with the ledger data without breaking the hash links. A hash of a most recently added blockchain block represents every entry on the chain that has come before it, making it possible to ensure that all peer nodes are in a consistent and trusted state. The chain may be stored on a peer node file system (i.e., local, attached storage, cloud, etc.), efficiently supporting the append-only nature of the blockchain workload.

[0043] The current state of the immutable ledger represents the latest values for all keys that are included in the chain entry log. Since the current state represents the latest key values known to a channel, it is sometimes referred to as a world state. Smart contract executable code invocations execute entries against the current state data of the ledger. To make these smart contract executable code interactions efficient, the latest values of the keys may be stored in a state database. The state database may be simply an indexed view into the chain's entry log and can therefore be regenerated from the chain at any time. The state database may automatically be recovered (or generated if needed) upon peer node startup and before entries are accepted.

[0044] A blockchain is different from a traditional database in that the blockchain is not a central storage but rather a decentralized, immutable, and secure storage, where nodes must share in changes to records in the storage. Some properties that are inherent in blockchain and which help implement the blockchain include, but are not limited to, an immutable ledger, smart contracts, security, privacy, decentralization, consensus, endorsement, accessibility, and the like.

[0045] Example embodiments provide a service to a particular vehicle and/or a user profile that is applied to the vehicle. For example, a user may be the owner of a vehicle or the operator of a vehicle owned by another party. The vehicle may require service at certain intervals, and the service needs may require authorization before permitting the services to be received. Also, service centers may offer services to vehicles in a nearby area based on the vehicle's current route plan and a relative level of service requirements (e.g., immediate, severe, intermediate, minor, etc.). The vehicle needs may be monitored via one or more vehicle and/or road sensors or cameras, which report sensed data to a central controller computer device in and/or apart from the vehicle. This data is forwarded to a management server for review and action. A sensor may be located on one or more of the interior of the transport, the exterior of the transport, on a fixed object apart from the transport, and on another transport proximate the transport. The sensor may also be associated with the transport's speed, the transport's brak-

ing, the transport's acceleration, fuel levels, service needs, the gear-shifting of the transport, the transport's steering, and the like. A sensor, as described herein, may also be a device, such as a wireless device in and/or proximate to the transport. Also, sensor information may be used to identify whether the vehicle is operating safely and whether an occupant has engaged in any unexpected vehicle conditions, such as during a vehicle access and/or utilization period. Vehicle information collected before, during and/or after a vehicle's operation may be identified and stored in a transaction on a shared/distributed ledger, which may be generated and committed to the immutable ledger as determined by a permission granting consortium, and thus in a "decentralized" manner, such as via a blockchain membership group.

[0046] Each interested party (i.e., owner, user, company, agency, etc.) may want to limit the exposure of private information, and therefore the blockchain and its immutability can be used to manage permissions for each particular user vehicle profile. A smart contract may be used to provide compensation, quantify a user profile score/rating/review, apply vehicle event permissions, determine when service is needed, identify a collision and/or degradation event, identify a safety concern event, identify parties to the event and provide distribution to registered entities seeking access to such vehicle event data. Also, the results may be identified, and the necessary information can be shared among the registered companies and/or individuals based on a consensus approach associated with the blockchain. Such an approach could not be implemented on a traditional centralized database.

[0047] Various driving systems of the instant solution can utilize software, an array of sensors as well as machine learning functionality, light detection and ranging (Lidar) projectors, radar, ultrasonic sensors, etc. to create a map of terrain and road that a transport can use for navigation and other purposes. In some embodiments, GPS, maps, cameras, sensors and the like can also be used in autonomous vehicles in place of Lidar.

[0048] The instant solution includes, in certain embodiments, authorizing a vehicle for service via an automated and quick authentication scheme. For example, driving up to a charging station or fuel pump may be performed by a vehicle operator or an autonomous transport and the authorization to receive charge or fuel may be performed without any delays provided the authorization is received by the service and/or charging station. A vehicle may provide a communication signal that provides an identification of a vehicle that has a currently active profile linked to an account that is authorized to accept a service, which can be later rectified by compensation. Additional measures may be used to provide further authentication, such as another identifier may be sent from the user's device wirelessly to the service center to replace or supplement the first authorization effort between the transport and the service center with an additional authorization effort.

[0049] Data shared and received may be stored in a database, which maintains data in one single database (e.g., database server) and generally at one particular location. This location is often a central computer, for example, a desktop central processing unit (CPU), a server CPU, or a mainframe computer. Information stored on a centralized database is typically accessible from multiple different points. A centralized database is easy to manage, maintain,

and control, especially for purposes of security because of its single location. Within a centralized database, data redundancy is minimized as a single storing place of all data also implies that a given set of data only has one primary record. A blockchain may be used for storing transport-related data and transactions.

[0050] Any of the actions described herein may be performed by one or more processors (such as a microprocessor, a sensor, an Electronic Control Unit (ECU), a head unit, and the like), with or without memory, which may be located on-board the transport and/or off-board the transport (such as a server, computer, mobile/wireless device, etc.). The one or more processors may communicate with other memory and/or other processors on-board or off-board other transports to utilize data being sent by and/or to the transport. The one or more processors and the other processors can send data, receive data, and utilize this data to perform one or more of the actions described or depicted herein.

[0051] FIG. 1A illustrates an example system to provide a decentralized power exchange 103. The current application may fully or partially execute on one or more of any vehicles depicted herein, a computer/server in the cloud/network, and any other processor that wirelessly communicates with the vehicles in the system, such as a mobile device. In the example of FIG. 1A, system 100 partially executes on cloud/network 101 and includes processor 102, which can execute instructions 106 stored on memory 104 to determine an inability for an entity to provide electricity over a threshold to an area, determine available energy from a group of electric vehicles from locations currently receiving energy in the area and a group of energy storage units in the area greater than the threshold, and provide the determined available energy to locations not currently receiving energy from the entity.

[0052] In reference to FIG. 1A, locations 110, 130, 150 are located in area 109. Locations can include any type of dwelling or structure (e.g., residence, office space, hotel, charging station, etc.). In the example of FIG. 1A, locations 110, 130, 150 are residences, but the current solution also contemplates other embodiments that include additional locations or other combinations of various types of locations within the area. The vehicles depicted in FIG. 1A can refer to any electric vehicle, hybrid vehicle, hydrogen fuel cell vehicle, plug-in hybrid vehicle, or any other type of vehicle that has a fuel cell stack, a motor, and/or a generator. Such vehicles can store electricity in rechargeable battery packs to be utilized for propelling the vehicle and, in some embodiments, for providing electricity back to a location (e.g., vehicle-to-home technology, vehicle-to-building technology, vehicle-to-grid technology, etc.) or for providing electricity to another vehicle (e.g., peer-to-peer vehicle charging, etc.).

[0053] In the example of FIG. 1A, location 110 includes vehicle 120, location 130 includes vehicle 140, and location 150 includes vehicle 160, but the current solution also contemplates other embodiments that include multiple vehicles at other types of locations, such as charging stations, etc. Vehicles 120, 140, 160 include processors 122, 142, 162, respectively, which may be referred to as Electronic Control Modules ("ECM"). Processors 122, 142, 162 can include networking technology to enable the exchange of data between one or more processors of the vehicles as well as one or more processors associated with the current solution via the Internet or other communications networks

(e.g., Bluetooth network, 5G wireless network, Wi-Fi, etc.). Vehicles 120, 140, 160 include memories 124, 144, 164, respectively, which can store instructions related to the current solution as well as other data specific to vehicles 120, 140, 160 (e.g., usage data, itinerary data, battery data, etc.). In addition, vehicles 120, 140, 160 include batteries 126, 146, 166, respectively, which can include rechargeable battery packs (e.g., lithium-ion battery cells, etc.) to store energy for propelling each respective vehicle, or for transferring the energy to another vehicle or location. Vehicles 120, 140, 160 also include connection devices 129, 149, 169, respectively, which can enable vehicle-to-vehicle connections (e.g., a power cord between connection device 129 of vehicle 120 and connection device 149 of vehicle 140, etc.) or vehicle-to-location connections (e.g., a power cord between connection device 149 of vehicle 140 and connection device 119 or location 110, etc.). Additionally, vehicles 120, 140, 160 include sensors 128, 148, 168, respectively, which can include GPS sensors, battery capacity sensors, camera sensors, as well as other sensors (e.g., radar, ultrasonic, LIDAR, etc.).

[0054] In reference to FIG. 1A, locations 110, 130, 150 also include power integration systems 111, 131, 151, respectively. Further, power integration systems 111, 131, 151 can include processors 112, 132, 152, respectively. In some embodiments, processors 112, 132, 152 can enable the distribution of electricity from the grid to any device that may operate at the respective locations (e.g., HVAC system, appliances, etc.). In other embodiments, processors 112, 132, 152 can disconnect the respective locations from the grid and enable the distribution of electricity from a power source other than the grid (e.g., a vehicle, solar panels, energy storage device, etc.) to any device that may operate at the respective locations. In addition, processors 112, 132, 152 can communicate both the real-time energy needs as well as historic energy patterns of the respective locations (stored on memories 114, 134, 154, respectively) to any other processor associated with the current solution. Power integration systems 111, 131, 151 can also include power integration mechanisms 116, 136, 156, respectively, to enable the distribution of electricity at the respective locations such as a distribution board (e.g., electric panel, breaker panel, etc.) to divide the electrical power feed into subsidiary circuits while providing a protective fuse or circuit breaker for each circuit in a common enclosure.

[0055] In the example of FIG. 1A, locations 110, 130, 150 also include energy storage units 113, 133, 153, respectively. Further, energy storage units 113, 133, 153 include processors 115, 135, 155, respectively, to enable communication with the other processors associated with the current solution. In addition, energy storage units 113, 133, 153 can also include batteries 117, 137, 157, respectively, to serve as a backup or supplemental power source for the respective locations (e.g., lithium-ion battery, lead-acid battery, etc.). Energy storage units 113, 133, 153 also include sensors 118, 138, 158 to detect battery capacity levels for batteries 117, 137, 157, respectively.

[0056] In further reference to the example of FIG. 1A, locations 130, 150 include energy generation systems 170, 180, respectively, to generate energy from renewable sources (e.g., sun, wind, etc.). Energy generation systems 170, 180 include processors 172, 182, respectively, to enable communication with the other processors associated with the current solution. In addition, energy generation systems 170,

180 include energy generation mechanisms **176**, **186**, respectively, to collect renewable resources and convert those renewable resources to electricity (e.g., solar panels, wind turbines, etc.).

[0057] Processors **112**, **115**, **122**, **132**, **135**, **142**, **152**, **155**, **162**, **172**, **182**, **192** or any other processor associated with a location, vehicle, or device related to the current solution may be referred to as Electronic Control Modules (“ECM”). In one example, the current solution depicted herein may occur wholly or partially in processor **102** or another processor associated with any location (e.g., processor **112**, processor **115**, processor **132**, processor **135**, etc.), such as an Electronic Control Unit (“ECU”), a computer in the infotainment system of the vehicle associated with the current solution (e.g., vehicle **120**, vehicle **140**, vehicle **160**, etc.), any device in the system (e.g., mobile device, tablet, etc.), and a computer or server which may be located outside of any location or vehicle depicted herein. Data obtained from any vehicle or system associated with the current solution (e.g., vehicle **120**, power integration system **131**, energy storage unit **153**, etc.), including data from a battery of such vehicles or systems (e.g., battery **126**, battery **137**, battery **166**, etc.) or data from a sensor of such vehicles or systems (e.g., sensors **128**, sensors **138**, sensors **168**, etc.) may be sent to processor **102** and processed therein. Data obtained from any vehicle or system associated with the current solution (e.g., vehicle **120**, power integration system **131**, energy storage unit **153**, etc.), including data from a battery of such vehicles or systems (e.g., battery **126**, battery **137**, battery **166**, etc.) or data from a sensor of such vehicles or systems (e.g., sensors **128**, sensors **138**, sensors **168**, etc.) may also be sent to a processor of any location associated with the current solution (e.g., processor **112**, processor **132**, processor **162**, etc.) and processed therein. The data, as well as the instructions related to the data, can be sent wirelessly (e.g., Bluetooth, wi-fi network, cellular network, etc.) between vehicles, locations, computers/servers in the cloud/network, and any other component related to the current solution.

[0058] In addition, data from processors **112**, **115**, **122**, **132**, **135**, **142**, **152**, **155**, **162**, **172**, **182**, **192** or any other processor associated with a location, vehicle, or device associated with the current solution may be sent to a computer/server in the cloud/network, processed therein and then stored in a database, which can maintain the data in many implementations, such as a single database (e.g., database server), along with other relevant data. For example, the data can be maintained at a central computer associated with the current solution, including processor **102**. Information stored on a centralized database is typically accessible from multiple different points. A centralized database is easy to manage, maintain, and control, especially for purposes of security because of its single location. Within a centralized database, data redundancy is minimized as a single storing place of all data also implies that a given set of data only has one primary record. In addition, blockchain may be used for storing vehicle-related data and transactions. In other embodiments, data from the vehicles may be stored in multiple computers or servers wherein the multiple computers or servers may be redundant.

[0059] Processor **102** can execute instructions **106** stored on memory **104** to cause system **100** to determine an inability for an entity to provide electricity over a threshold to a location in an area. In the example of FIG. 1A, system

100 determines that the entity, electric grid **190**, is unable to provide electricity over the threshold to location **110**, located in area **109**. In some embodiments, the threshold relates to an amount of electricity required to operate the devices utilized at location **110** (e.g., HVAC system, appliances, vehicle charger, etc.). In such embodiments, processor **102** can communicate with processor **112** of power integration system **111** to analyze real-time power usage data as well as historical power usage data stored on memory **114** to determine the amount of electricity required (or a range of the amount of electricity required) to operate the devices utilized at location **110** and to define the amount threshold (e.g., 30 kWh per day, etc.). Further, when a total loss of power occurs (e.g., blackout) or where a partial loss of power occurs (e.g., brownout, etc.), system **100** can determine whether grid **190** has the ability to provide electricity over the amount threshold by communicating with processor **192** of grid **190** and/or processor **112** of power integration system **111** and comparing the amount of incoming power at location **110** to the amount threshold set at location **110**. If the amount of incoming power at location **110** (or anticipated amount of incoming power at location **100** based on brown out alerts, for example, from processor **192** of grid **190**) is below the amount threshold, then system **100** can determine that the entity has an inability to provide electricity above the amount threshold. In this way, the current solution defines what is necessary for locations to remain fully operational during a blackout and to prevent potential damage or other functional impairment of the devices operating during moments of partial losses of power, such as brownouts.

[0060] In other embodiments, the threshold relates to a cost or price of electricity associated with powering a location, where processor **102** can execute instructions **106** stored on memory **104** to cause system **100** to determine an inability for an entity to provide electricity under a threshold to the location in the area. In such embodiments, processor **102** can communicate with processor **112** of power integration system **111** to analyze real-time power usage data as well as historical power usage data stored on memory **114** and the associated cost to determine the cost of electricity required (or a range of the cost of electricity required) to operate the devices utilized at location **110** and to define a price threshold (e.g., \$10 per day, etc.). Further, at a particular time of day (e.g., peak hours, etc.), system **100** can determine whether grid **190** has the ability to provide electricity under the price threshold by communicating with processor **192** of grid **190** and/or processor **112** of power integration system **111** and comparing the price of incoming power at location **110** to the price threshold set at location **110**. If the price of incoming power at location **110** is above the price threshold, then system **100** can determine that the entity has an inability to provide electricity under the threshold. In this way, the current solution enables the location to seek out other sources of electricity that are more cost beneficial to the location. In other embodiments still, the current solution can include concurrent determinations for both an inability for an entity to provide electricity over an amount threshold as well as an inability for an entity to provide electricity under a price threshold.

[0061] Where the threshold relates to an amount of electricity, processor **102** can execute instructions **106** stored on memory **104** to cause system **100** to determine available energy from a group of electric vehicles from locations in

the area and a group of energy storage units in the area greater than the threshold. In some embodiments, the locations can be currently receiving energy in the area (e.g., from grid 190, etc.), which can contribute to the available energy from the group of vehicles or the group of energy storage units provided the group of vehicles and/or the energy storage units are coupled to the energy source (e.g., connection device 149 of vehicle 140 coupled to connection device 139 of power integration system 131, etc.). In other embodiments, the locations may not be currently receiving energy from the area but may still have available energy being stored at the location (e.g., energy storage unit 133 at location 130, etc.) or being generated from “off grid” sources (e.g., energy generation system 170 at location 130, etc.).

[0062] The group of electric vehicles can be located at the same location (e.g., fleet of electric vehicles, etc.) or at disparate locations (e.g., individual residences, etc.). In the example of FIG. 1A, the group of vehicles includes vehicle 140 located at location 130 and vehicle 160 located at location 150. When grid 190 is unable to provide electricity over an amount threshold to location 110 as described herein, processor 102 can communicate with the processors of the group of vehicles (e.g., processor 142 of vehicle 140, processor 162 of vehicle 160, etc.), and, based on data from the sensors of the group of vehicles (e.g., sensors 148 of vehicle 140, sensors 168 of vehicle 160, etc.) system 100 can determine the available energy stored on the batteries of the group of vehicles (e.g., battery 146 of vehicle 140, battery 166 of vehicle 160, etc.).

[0063] Further, the group of energy storage units can be located at the same location (e.g., charging station, etc.) or at disparate locations (e.g., individual residences, etc.). In the example of FIG. 1A, the group of energy storage units includes energy storage unit 133 located at location 130 and energy storage unit 153 located at location 150. When grid 190 is unable to provide electricity over an amount threshold to location 110 as described herein, processor 102 can communicate with the processors of the group of energy storage units (e.g., processor 135 of energy storage unit 133, processor 155 of energy storage unit 153, etc.), and, based on data from the sensors of the group of energy storage units (e.g., sensors 138 of energy storage unit 133, sensors 158 of energy storage unit 153, etc.) system 100 can determine the available energy stored on the batteries of the group of energy storage units (e.g., battery 137 of energy storage unit 133, battery 157 of energy storage unit 153, etc.).

[0064] The availability of energy from the group of electric vehicles and the group of energy storage units can be affected by several factors. In some embodiments, the availability of energy from the group of electric vehicles and the group of energy storage units can also include an anticipated surplus not currently realized at the location. For example, processor 102 can communicate with processors 132, 135, 142 at location 130 to determine that the current available energy from location 130 is 50 kWh (e.g., 40 kWh from battery 146 of vehicle 140 plus 10 kWh from battery 137 of energy storage unit 133). However, based on historical data stored in memory 134 of power integration device 131, data from sensors 138 related to the battery capacity of battery 137 of energy storage unit 133, and real-time weather data specific to location 130 collected by processor 102, system 100 can determine that energy generation system 170 will generate an additional 5 kWh of energy at location 130 that cannot be stored (e.g., battery 137 at capacity, etc.). In such

instances, processor 102 can communicate with processor 132 of power integration system 131 to cause energy storage unit 133 to transfer its energy from battery 137 to battery 146 of vehicle 140, so that, in effect, vehicle 140 receives its full charge from energy storage unit 133, rather than from grid 190. In this way, battery 137 of energy storage unit 133 can free up storage space in order to gain the capacity needed to capture the anticipated surplus when it arises. As a result, whereas previously only 40 kWh may have been available if vehicle 140 was determined to be the delivery mechanism of the available energy, now, after the transfer of 10 kWh from battery 137 of energy storage unit 133 to battery 146 of vehicle 140 in order to “make room” for the anticipated surplus, vehicle 140 can deliver 50 kWh of available energy to a location.

[0065] In addition, the availability of energy from the group of electric vehicles and the group of energy storage units can also include the expenditure of energy from traveling back and forth between locations. For example, processor 102 can analyze GPS data of location 110 and location 150 in order to determine that the distance between location 110 and location 150 is 5 miles, totaling a 10-mile round trip between the locations. Further, processor 102 can communicate with processor 162 of vehicle 160 to analyze the usage data stored on memory 164 of vehicle 160 to determine the kWh/mile rate for vehicle 160. If, for example, vehicle 160 utilizes 0.5 kWh/mile, then the 10-mile round trip will expend approximately 5 kWh. So, if system 100 initially determines that vehicle 160 can deliver 40 kWh of available energy, then, based on the estimated 5 kWh expended traveling between locations, the adjusted available energy is 35 kWh (e.g., 40 kWh of available energy minus 5 kWh expended traveling between locations). Further, additional reductions in the available energy can include limits on the available energy for a particular location. For example, while 35 kWh of energy may be available from location 150 (e.g., 40 kWh of available energy minus 5 kWh expended traveling between locations), it may be determined that only 20 kWh can be provided to another location (e.g., location 110) so that the remaining 15 kWh (35 kWh available minus 20 kWh made available) can be kept in reserve for location 150. These restrictions on the amount of energy that can be made available can be set by a user manually (e.g., through a user interface of power integration system 151, etc.) or automatically by system 100, which can determine the amount of energy to be made available based on processor 102 communicating with a processor of the location (e.g., processor 152 of power integration system 151, etc.) and analyzing energy usage data stored on a memory of a power integration system 151 (e.g., memory 154 of power integration system 151, etc.).

[0066] Where the threshold relates to a price of electricity (or includes the price in addition to determinations related to the amount of electricity threshold), processor 102 can execute instructions 106 stored on memory 104 to cause system 100 to determine available energy from a group of electric vehicles from locations currently receiving energy in the area and a group of energy storage units in the area less than the threshold. System 100 can determine a price range for both the location seeking the available energy and the location(s) potentially providing the available energy based on the sale and buyback prices offered by the grid. For example, in reference to FIG. 1A, processor 102 can communicate with processor 192 to determine that grid 190 is

currently selling electricity for \$0.30/kWh but buying back electricity (e.g., net metering, etc.) for \$0.10/kWh. In such instances, processor 102 can communicate with the processors of the locations seeking to purchase electricity (e.g., processor 112 of power integration system 111 at location 110), and set the price threshold to purchase electricity to be below \$0.30/kWh. Further, processor 102 can communicate with the processors of the locations offering to sell their electricity (e.g., processors 132, 142 at location 130, processors 152, 162 at location 150, etc.), and set the price threshold to sell their electricity above \$0.10/kWh. In this way, system 100 can act as a broker to secure a beneficial transaction for location 110, which would purchase electricity from location 130 for less than it would from grid 190, as well as a beneficial transaction for location 130, which would sell its electricity to location 110 for more than it would to grid 190.

[0067] Processor 102 can execute instructions 106 stored on memory 104 to cause system 100 to provide the determined available energy to locations not currently receiving energy from the entity. In reference to the example of FIG. 1A, system 100 determines that grid 190 has an inability to provide electricity to location 110 over the amount threshold of 50 kWh and below the price threshold of \$0.30/kWh. As a result, processor 102 communicates with the processors of location 130 (e.g., processor 132, processor 142, etc.) and the processors of location 150 (e.g., processor 152, processor 162, etc.). System 100 can determine that location 130 has 60 kWh of available energy stored on battery 146 of vehicle 140 for a price of \$0.20/kWh, and that location 150 has 40 kWh of available energy stored on battery 166 of vehicle 160 for a price of \$0.12/kWh. However, based on historical data stored in memory 154 of power integration system 151, data from sensors 158 related to the battery capacity of battery 157 of energy storage unit 153, and real-time weather data specific to location 150 collected by processor 102, system 100 can determine that energy generation system 180 will be able to generate an additional 10 kWh of energy that location 150 will not be able to store (e.g., battery 157 at capacity, etc.). In such instances, system 100 can cause the energy stored in battery 157 of energy storage unit 153 (which had not previously been made available for sale) to be transferred to battery 166 of vehicle 160. If, for example, battery 157 of energy storage unit 153 had 10 kWh of stored energy, then, after the transfer, location 150 would now have 50 kWh of available energy stored on battery 166 of vehicle 160 for a price of \$0.12/kWh (40 kWh initially stored on battery 166 of vehicle 160 plus the 10 kWh of energy transferred from battery 157 of energy storage unit 153). Since the available energy at both location 130 and location 150 is now greater than the amount threshold set by location 110, and since the price of energy at both location 130 and location 150 is below the price threshold set by location 110, system 100 can cause location 150 to provide the determined amount of available energy to location 110 since it offers the same amount of energy but for a lower cost.

[0068] In some embodiments, providing the determined available energy includes dispatching a vehicle from the group of electric vehicles to locations not currently receiving energy (and/or requesting energy at a cheaper price). When dispatching a vehicle, processor 102 can send a notification to a processor of the vehicle (e.g., processor 122 of vehicle 130, processor 162 of vehicle 160, etc.), where the notifi-

cation can include instructions 106 to be stored on a memory of the vehicle (e.g., memory 124 of vehicle 120, memory 164 of vehicle 160, etc.) that can be executed to cause the vehicle to travel autonomously between locations. In other examples, the notification can include instructions 106 to be stored on a memory of the vehicle (e.g., memory 124 of vehicle 120, memory 164 of vehicle 160, etc.) that can provide directions to travel manually between locations. FIG. 1B illustrates an example system to provide a decentralized power exchange by dispatching a vehicle to a location not currently receiving energy 105. In the example of FIG. 1B, vehicle 160 is dispatched from location 150 to location 110. In some examples, vehicle 160 can connect to vehicle 120 (e.g., power cord from connection device 169 of vehicle 160 to connection device 129 of vehicle 120, etc.) and transfer the 50 kWh of available energy from battery 166 of vehicle 160 to battery 126 of vehicle 120 (e.g., peer-to-peer charging). In the example of FIG. 1B, a power cord from connection device 169 of vehicle 160 connects to connection device 119 of power integration system 111 to facilitate the transfer of the 50 kWh of available energy from battery 166 of vehicle 160 to power integration system 111 where it can be used instantaneously to power location 110, or where some or all of the 50 kWh can be diverted to battery 117 of energy storage unit 113.

[0069] In other embodiments, providing the determined available energy includes dispatching a vehicle from locations not currently receiving energy (and/or requesting energy at a cheaper price) to the group of energy storage units. FIG. 1C illustrates an example system to provide a decentralized power exchange by dispatching a vehicle from a location not currently receiving energy 107. In the example of FIG. 1C, vehicle 120 is dispatched from location 110 to location 150. In some examples, vehicle 120 can connect to power integration system 151 (e.g., power cord from connection device 129 of vehicle 120 to connection device 159 of power integration system 151, etc.) and transfer the available energy from battery 157 of energy storage unit 153 to battery 126 of vehicle 120. In the example of FIG. 1B, a power cord from connection device 169 of vehicle 160 connects to connection device 129 of vehicle 120 to facilitate the transfer of the 50 kWh of available energy from battery 166 of vehicle 160 to battery 126 of vehicle 120, where it can be transported back to location 110 to be utilized as its power source.

[0070] Flow diagrams depicted herein, such as FIG. 1A, FIG. 1B, FIG. 1C, FIG. 2A, FIG. 2B, FIG. 2C, FIG. 2D, FIG. 2E, FIG. 3A, FIG. 3B, FIG. 3C, FIG. 4, FIG. 5A, FIG. 5B, FIG. 6A, FIG. 6B, FIG. 6C, FIG. 6D, and FIG. 7 are separate examples but may be the same or different embodiments. Any of the operations in one flow diagram could be adopted and shared with another flow diagram. No example operation is intended to limit the subject matter of any embodiment or corresponding claim.

[0071] It is important to note that all the flow diagrams and corresponding processes derived from FIG. 1A, FIG. 1B, FIG. 1C, FIG. 2A, FIG. 2B, FIG. 2C, FIG. 2D, FIG. 2E, FIG. 3A, FIG. 3B, FIG. 3C, FIG. 4, FIG. 5A, FIG. 5B, FIG. 6A, FIG. 6B, FIG. 6C, FIG. 6D, and FIG. 7 may be part of a same process or may share sub-processes with one another thus making the diagrams combinable into a single preferred embodiment that does not require any one specific operation but which performs certain operations from one example process and from one or more additional processes. All the

example processes are related to the same physical system and can be used separately or interchangeably.

[0072] The instant solution can be used in conjunction with one or more types of vehicles: battery electric vehicles, hybrid vehicles, fuel cell vehicles, internal combustion engine vehicles and/or vehicles utilizing renewable sources.

[0073] FIG. 2A illustrates a transport network diagram 200, according to example embodiments. The network comprises elements including a transport 202 including a processor 204, as well as a transport 202' including a processor 204'. The transports 202, 202' communicate with one another via the processors 204, 204', as well as other elements (not shown) including transceivers, transmitters, receivers, storage, sensors, and other elements capable of providing communication. The communication between the transports 202, and 202' can occur directly, via a private and/or a public network (not shown), or via other transports and elements comprising one or more of a processor, memory, and software. Although depicted as single transports and processors, a plurality of transports and processors may be present. One or more of the applications, features, steps, solutions, etc., described and/or depicted herein may be utilized and/or provided by the instant elements.

[0074] FIG. 2B illustrates another transport network diagram 210, according to example embodiments. The network comprises elements including a transport 202 including a processor 204, as well as a transport 202' including a processor 204'. The transports 202, 202' communicate with one another via the processors 204, 204', as well as other elements (not shown), including transceivers, transmitters, receivers, storage, sensors, and other elements capable of providing communication. The communication between the transports 202, and 202' can occur directly, via a private and/or a public network (not shown), or via other transports and elements comprising one or more of a processor, memory, and software. The processors 204, 204' can further communicate with one or more elements 230 including sensor 212, wired device 214, wireless device 216, database 218, mobile phone 220, transport 222, computer 224, I/O device 226, and voice application 228. The processors 204, 204' can further communicate with elements comprising one or more of a processor, memory, and software.

[0075] Although depicted as single transports, processors and elements, a plurality of transports, processors and elements may be present. Information or communication can occur to and/or from any of the processors 204, 204' and elements 230. For example, the mobile phone 220 may provide information to the processor 204, which may initiate the transport 202 to take an action, may further provide the information or additional information to the processor 204', which may initiate the transport 202' to take an action, may further provide the information or additional information to the mobile phone 220, the transport 222, and/or the computer 224. One or more of the applications, features, steps, solutions, etc., described and/or depicted herein may be utilized and/or provided by the instant elements.

[0076] FIG. 2C illustrates yet another transport network diagram 240, according to example embodiments. The network comprises elements including a transport 202, a processor 204, and a non-transitory computer readable medium 242C. The processor 204 is communicably coupled to the computer readable medium 242C and elements 230 (which

were depicted in FIG. 2B). The transport 202 could be a transport, server, or any device with a processor and memory.

[0077] The processor 204 performs one or more of the current solution includes one or more of determining an inability for an entity to provide electricity over a threshold to an area 244C; determining available energy from a group of electric vehicles from locations currently receiving energy in the area and a group of energy storage units in the area greater than the threshold 246C; providing the determined available energy to locations not currently receiving energy from the entity 248C.

[0078] FIG. 2D illustrates a further transport network diagram 250, according to example embodiments. The network comprises elements including a transport 202 a processor 204, and a non-transitory computer readable medium 242D. The processor 204 is communicably coupled to the computer readable medium 242D and elements 230 (which were depicted in FIG. 2B). The transport 202 could be a transport, server or any device with a processor and memory.

[0079] The processor 204 performs one or more of wherein the threshold relates to an amount of electricity required to power a first location in the area 244D; wherein the threshold relates to a cost of electricity associated with powering a first location in the area 245D; wherein determining available energy from the group of electric vehicles and the group of energy storage units includes predicting an anticipated surplus of available energy 246D; wherein determining available energy from the group of electric vehicles and the group of energy storage units includes energy expended traveling between the locations not currently receiving energy from the entity and the locations of the group of vehicles and the group of energy storage units 247D; wherein providing the determined available energy includes dispatching a vehicle from the group of electric vehicles to locations not currently receiving energy 248D; wherein providing the determined available energy includes dispatching a vehicle from locations not currently receiving energy to the group of energy storage units 249D.

[0080] FIG. 2E illustrates yet a further transport network diagram 260, according to example embodiments. Referring to FIG. 2E, the network diagram 260 includes a transport 202 connected to other transports 202' and to an update server node 203 over a blockchain network 206. The transports 202 and 202' may represent transports/vehicles. The blockchain network 206 may have a ledger 208 for storing software update validation data and a source 207 of the validation for future use (e.g., for an audit).

[0081] While this example describes in detail only one transport 202, multiple such nodes may be connected to the blockchain 206. It should be understood that the transport 202 may include additional components and that some of the components described herein may be removed and/or modified without departing from a scope of the instant application. The transport 202 may have a computing device or a server computer, or the like, and may include a processor 204, which may be a semiconductor-based microprocessor, a central processing unit (CPU), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), and/or another hardware device. Although a single processor 204 is depicted, it should be understood that the transport 202 may include multiple processors, multiple cores, or the like without departing from the scope of the

instant application. The transport **202** could be a transport, server or any device with a processor and memory.

[0082] The processor **204** performs one or more of receiving a confirmation of an event from one or more elements described or depicted herein, wherein the confirmation comprises a blockchain consensus between peers represented by any of the elements **244E** and executing a smart contract to record the confirmation on a blockchain-based on the blockchain consensus **246E**. Consensus is formed between one or more of any element **230** and/or any element described or depicted herein, including a transport, a server, a wireless device, etc. In another example, the transport **202** can be one or more of any element **230** and/or any element described or depicted herein, including a server, a wireless device, etc.

[0083] The processors and/or computer readable medium **242E** may fully or partially reside in the interior or exterior of the transports. The steps or features stored in the computer readable medium **242E** may be fully or partially performed by any of the processors and/or elements in any order. Additionally, one or more steps or features may be added, omitted, combined, performed at a later time, etc.

[0084] FIG. 2F illustrates a diagram **265** depicting the electrification of one or more elements. In one example, a transport **266** may provide power stored in its batteries to one or more elements, including other transport(s) **268**, charging station(s) **270**, and electric grid(s) **272**. The electric grid(s) **272** is/are coupled to one or more of the charging stations **270**, which may be coupled to one or more of the transports **268**. This configuration allows the distribution of electricity/power received from the transport **266**. The transport **266** may also interact with the other transport(s) **268**, such as via Vehicle to Vehicle (V2V) technology, communication over cellular, WiFi, and the like. The transport **266** may also interact wirelessly and/or wired with other transports **268**, the charging station(s) **270** and/or with the electric grid(s) **272**. In one example, the transport **266** is routed (or routes itself) in a safe and efficient manner to the electric grid(s) **272**, the charging station(s) **270**, or the other transport(s) **268**. Using one or more embodiments of the instant solution, the transport **266** can provide energy to one or more of the elements depicted herein in various advantageous ways as described and/or depicted herein. Further, the safety and efficiency of the transport may be increased, and the environment may be positively affected as described and/or depicted herein.

[0085] The term ‘energy’, ‘electricity’, ‘power’, and the like may be used to denote any form of energy received, stored, used, shared, and/or lost by the vehicles(s). The energy may be referred to in conjunction with a voltage source and/or a current supply of charge provided from an entity to the transport(s) during a charge/use operation. Energy may also be in the form of fossil fuels (for example, for use with a hybrid transport) or via alternative power sources, including but not limited to lithium-based, nickel-based, hydrogen fuel cells, atomic/nuclear energy, fusion-based energy sources, and energy generated on-the-fly during an energy sharing and/or usage operation for increasing or decreasing one or more transports energy levels at a given time.

[0086] In one example, the charging station **270** manages the amount of energy transferred from the transport **266** such that there is sufficient charge remaining in the transport **266** to arrive at a destination. In one example, a wireless connection is used to wirelessly direct an amount of energy

transfer between transports **268**, wherein the transports may both be in motion. In one embodiment, wireless charging may occur via a fixed charger and batteries of the transport in alignment with one another (such as a charging mat in a garage or parking space). In one example, an idle vehicle, such as a vehicle **266** (which may be autonomous) is directed to provide an amount of energy to a charging station **270** and return to the original location (for example, its original location or a different destination). In one example, a mobile energy storage unit (not shown) is used to collect surplus energy from at least one other transport **268** and transfer the stored surplus energy at a charging station **270**. In one example, factors determine an amount of energy to transfer to a charging station **270**, such as distance, time, as well as traffic conditions, road conditions, environmental/weather conditions, the vehicle’s condition (weight, etc.), an occupant(s) schedule while utilizing the vehicle, a prospective occupant(s) schedule waiting for the vehicle, etc. In one example, the transport(s) **268**, the charging station(s) **270** and/or the electric grid(s) **272** can provide energy to the transport **266**.

[0087] In one embodiment, a location such as a building, a residence, or the like (not depicted), communicably coupled to one or more of the electric grid **272**, the transport **266**, and/or the charging station(s) **270**. The rate of electric flow to one or more of the location, the transport **266**, the other transport(s) **268** is modified, depending on external conditions, such as weather. For example, when the external temperature is extremely hot or extremely cold, raising the chance for an outage of electricity, the flow of electricity to a connected vehicle **266/268** is slowed to help minimize the chance for an outage.

[0088] In one embodiment, transports **266** and **268** may be utilized as bidirectional transports. Bidirectional transports are those that may serve as mobile microgrids that can assist in the supplying of electrical power to the grid **272** and/or reduce the power consumption when the grid is stressed. Bidirectional transports incorporate bidirectional charging, which in addition to receiving a charge to the transport, the transport can take energy from the transport and “push” the energy back into the grid **272**, otherwise referred to as “V2G”. In bidirectional charging, the electricity flows both ways; to the transport and from the transport. When a transport is charged, alternating current (AC) electricity from the grid **272** is converted to direct current (DC). This may be performed by one or more of the transport’s own converter or a converter on the charger **270**. The energy stored in the transport’s batteries may be sent in an opposite direction back to the grid. The energy is converted from DC to AC through a converter usually located in the charger **270**, otherwise referred to as a bidirectional charger. Further, the instant solution as described and depicted with respect to FIG. 2F can be utilized in this and other networks and/or systems.

[0089] FIG. 2G is a diagram showing interconnections between different elements **275**. The instant solution may be stored and/or executed entirely or partially on and/or by one or more computing devices **278'**, **279'**, **281'**, **282'**, **283'**, **284'**, **276'**, **285'**, **287'** and **277'** associated with various entities, all communicably coupled and in communication with a network **286**. A database **287** is communicably coupled to the network and allows for the storage and retrieval of data. In one example, the database is an immutable ledger. One or more of the various entities may be a transport **276**, one or

more service provider **279**, one or more public buildings **281**, one or more traffic infrastructure **282**, one or more residential dwellings **283**, an electric grid/charging station **284**, a microphone **285**, and/or another transport **277**. Other entities and/or devices, such as one or more private users using a smartphone **278**, a laptop **280**, an augmented reality (AR) device, a virtual reality (VR) device, and/or any wearable device may also interwork with the instant solution. The smartphone **278**, laptop **280**, the microphone **285**, and other devices may be connected to one or more of the connected computing devices **278'**, **279'**, **281'**, **282'**, **283'**, **284'**, **276'**, **285'**, **287'**, and **277'**. The one or more public buildings **281** may include various agencies. The one or more public buildings **281** may utilize a computing device **281'**. The one or more service provider **279** may include a dealership, a tow truck service, a collision center or other repair shop. The one or more service provider **279** may utilize a computing apparatus **279'**. These various computer devices may be directly and/or communicably coupled to one another, such as via wired networks, wireless networks, blockchain networks, and the like. The microphone **285** may be utilized as a virtual assistant, in one example. In one example, the one or more traffic infrastructure **282** may include one or more traffic signals, one or more sensors including one or more cameras, vehicle speed sensors or traffic sensors, and/or other traffic infrastructure. The one or more traffic infrastructure **282** may utilize a computing device **282'**.

[0090] In one embodiment, anytime an electrical charge is given or received to/from a charging station and/or an electrical grid, the entities that allow that to occur are one or more of a vehicle, a charging station, a server, and a network communicably coupled to the vehicle, the charging station, and the electrical grid.

[0091] In one example, a transport **277/276** can transport a person, an object, a permanently or temporarily affixed apparatus, and the like. In one example, the transport **277** may communicate with transport **276** via V2V communication through the computers associated with each transport **276'** and **277'** and may be referred to as a transport, car, vehicle, automobile, and the like. The transport **276/277** may be a self-propelled wheeled conveyance, such as a car, a sports utility vehicle, a truck, a bus, a van, or other motor or battery-driven or fuel cell-driven transport. For example, transport **276/277** may be an electric vehicle, a hybrid vehicle, a hydrogen fuel cell vehicle, a plug-in hybrid vehicle, or any other type of vehicle with a fuel cell stack, a motor, and/or a generator. Other examples of vehicles include bicycles, scooters, trains, planes, boats, and any other form of conveyance that is capable of transportation. The transport **276/277** may be semi-autonomous or autonomous. For example, transport **276/277** may be self-manuevering and navigate without human input. An autonomous vehicle may have and use one or more sensors and/or a navigation unit to drive autonomously.

[0092] FIG. 2H is another block diagram showing interconnections between different elements in one example **290**. A transport **276** is presented and includes ECUs **295**, **296**, and a Head Unit (otherwise known as an Infotainment System) **297**. An Electrical Control Unit (ECU) is an embedded system in automotive electronics controlling one or more of the electrical systems or subsystems in a transport. ECUs may include but are not limited to the management of a transport's engine, brake system, gearbox system, door

locks, dashboard, airbag system, infotainment system, electronic differential, and active suspension. ECUs are connected to the transport's Controller Area Network (CAN) bus **294**. The ECUs may also communicate with a transport computer **298** via the CAN bus **294**. The transport's processors/sensors (such as the transport computer) **298** can communicate with external elements, such as a server **293** via a network **292** (such as the Internet). Each ECU **295**, **296**, and Head Unit **297** may contain its own security policy. The security policy defines permissible processes that can be executed in the proper context. In one example, the security policy may be partially or entirely provided in the transport computer **298**.

[0093] ECUs **295**, **296**, and Head Unit **297** may each include a custom security functionality element **299** defining authorized processes and contexts within which those processes are permitted to run. Context-based authorization to determine validity if a process can be executed allows ECUs to maintain secure operation and prevent unauthorized access from elements such as the transport's Controller Area Network (CAN Bus). When an ECU encounters a process that is unauthorized, that ECU can block the process from operating. Automotive ECUs can use different contexts to determine whether a process is operating within its permitted bounds, such as proximity contexts such as nearby objects, distance to approaching objects, speed, and trajectory relative to other moving objects, and operational contexts such as an indication of whether the transport is moving or parked, the transport's current speed, the transmission state, user-related contexts such as devices connected to the transport via wireless protocols, use of the infotainment, cruise control, parking assist, driving assist, location-based contexts, and/or other contexts.

[0094] Referring to FIG. 2I, an operating environment **290A** for a connected transport, is illustrated according to some embodiments. As depicted, the transport **276** includes a Controller Area Network (CAN) bus **291A** connecting elements **292A-299A** of the transport. Other elements may be connected to the CAN bus and are not depicted herein. The depicted elements connected to the CAN bus include a sensor set **292A**, Electronic Control Units **293A**, autonomous features or Advanced Driver Assistance Systems (ADAS) **294A**, and the navigation system **295A**. In some embodiments, the transport **276** includes a processor **296A**, a memory **297A**, a communication unit **298A**, and an electronic display **299A**.

[0095] The processor **296A** includes an arithmetic logic unit, a microprocessor, a general-purpose controller, and/or a similar processor array to perform computations and provide electronic display signals to a display unit **299A**. The processor **296A** processes data signals and may include various computing architectures, including a complex instruction set computer (CISC) architecture, a reduced instruction set computer (RISC) architecture, or an architecture implementing a combination of instruction sets. The transport **276** may include one or more processors **296A**. Other processors, operating systems, sensors, displays, and physical configurations that are communicably coupled to one another (not depicted) may be used with the instant solution.

[0096] Memory **297A** is a non-transitory memory storing instructions or data that may be accessed and executed by the processor **296A**. The instructions and/or data may include code to perform the techniques described herein.

The memory **297A** may be a dynamic random-access memory (DRAM) device, a static random-access memory (SRAM) device, flash memory, or another memory device. In some embodiments, the memory **297A** also may include non-volatile memory or a similar permanent storage device and media, which may include a hard disk drive, a floppy disk drive, a CD-ROM device, a DVD-ROM device, a DVD-RAM device, a DVD-RW device, a flash memory device, or some other mass storage device for storing information on a permanent basis. A portion of the memory **297A** may be reserved for use as a buffer or virtual random-access memory (virtual RAM). The transport **276** may include one or more memories **297A** without deviating from the current solution.

[0097] The memory **297A** of the transport **276** may store one or more of the following types of data: navigation route data **295A**, and autonomous features data **294A**. In some embodiments, the memory **297A** stores data that may be necessary for the navigation application **295A** to provide the functions.

[0098] The navigation system **295A** may describe at least one navigation route including a start point and an endpoint. In some embodiments, the navigation system **295A** of the transport **276** receives a request from a user for navigation routes wherein the request includes a starting point and an ending point. The navigation system **295A** may query a real-time data server **293** (via a network **292**), such as a server that provides driving directions, for navigation route data corresponding to navigation routes, including the start point and the endpoint. The real-time data server **293** transmits the navigation route data to the transport **276** via a wireless network **292**, and the communication system **298A** stores the navigation data **295A** in the memory **297A** of the transport **276**.

[0099] The ECU **293A** controls the operation of many of the systems of the transport **276**, including the ADAS systems **294A**. The ECU **293A** may, responsive to instructions received from the navigation system **295A**, deactivate any unsafe and/or unselected autonomous features for the duration of a journey controlled by the ADAS systems **294A**. In this way, the navigation system **295A** may control whether ADAS systems **294A** are activated or enabled so that they may be activated for a given navigation route.

[0100] The sensor set **292A** may include any sensors in the transport **276** generating sensor data. For example, the sensor set **292A** may include short-range sensors and long-range sensors. In some embodiments, the sensor set **292A** of the transport **276** may include one or more of the following vehicle sensors: a camera, a Lidar sensor, an ultrasonic sensor, an automobile engine sensor, a radar sensor, a laser altimeter, a manifold absolute pressure sensor, an infrared detector, a motion detector, a thermostat, a sound detector, a carbon monoxide sensor, a carbon dioxide sensor, an oxygen sensor, a mass airflow sensor, an engine coolant temperature sensor, a throttle position sensor, a crankshaft position sensor, a valve timer, an air-fuel ratio meter, a blind spot meter, a curb feeler, a defect detector, a Hall effect sensor, a parking sensor, a radar gun, a speedometer, a speed sensor, a tire-pressure monitoring sensor, a torque sensor, a transmission fluid temperature sensor, a turbine speed sensor (TSS), a variable reluctance sensor, a vehicle speed sensor (VSS), a water sensor, a wheel speed sensor, a GPS sensor,

a mapping functionality, and any other type of automotive sensor. The navigation system **295A** may store the sensor data in the memory **297A**.

[0101] The communication unit **298A** transmits and receives data to and from the network **292** or to another communication channel. In some embodiments, the communication unit **298A** may include a DSRC transceiver, a DSRC receiver, and other hardware or software necessary to make the transport **276** a DSRC-equipped device.

[0102] The transport **276** may interact with other transports **277** via V2V technology. V2V communication includes sensing radar information corresponding to relative distances to external objects, receiving GPS information of the transports, setting areas as areas where the other transports **277** are located based on the sensed radar information, calculating probabilities that the GPS information of the object vehicles will be located at the set areas, and identifying transports and/or objects corresponding to the radar information and the GPS information of the object vehicles based on the calculated probabilities, in one example.

[0103] For a transport to be adequately secured, the transport must be protected from unauthorized physical access as well as unauthorized remote access (e.g., cyber-threats). To prevent unauthorized physical access, a transport is equipped with a secure access system such as a keyless entry in one example. Meanwhile, security protocols are added to a transport's computers and computer networks to facilitate secure remote communications to and from the transport in one example.

[0104] Electronic Control Units (ECUs) are nodes within a transport that control tasks such as activating the windshield wipers to tasks such as an anti-lock brake system. ECUs are often connected to one another through the transport's central network, which may be referred to as a controller area network (CAN). State-of-the-art features such as autonomous driving are strongly reliant on implementing new, complex ECUs such as advanced driver-assistance systems (ADAS), sensors, and the like. While these new technologies have helped improve the safety and driving experience of a transport, they have also increased the number of externally-communicating units inside of the transport, making them more vulnerable to attack. Below are some examples of protecting the transport from physical intrusion and remote intrusion.

[0105] FIG. 2J illustrates a keyless entry system **290B** to prevent unauthorized physical access to a transport **291B**, according to example embodiments. Referring to FIG. 2J, a key fob **292B** transmits commands to a transport **291B** using radio frequency signals in one example. In this example, the key fob **292B** includes a transmitter **2921B** with an antenna that is capable of sending short-range wireless radio signals. The transport **291B** includes a receiver **2911B** with an antenna that is capable of receiving the short-range wireless signal transmitted from the transmitter **2921B**. The key fob **292B** and the transport **291B** also include CPUs **2922B** and **2913B**, respectively, which control the respective devices. Here, a memory of the CPUs **2922B** and **2913B** (or accessible to the CPUs). Each of the key fob **292B** and the transport **291B** includes power supplies **2924B** and **2915B** for powering the respective devices in one example.

[0106] When the user presses a button **293B** (or otherwise actuates the fob, etc.) on the key fob **292B**, the CPU **2922B** wakes up inside the key fob **292B** and sends a data stream to the transmitter **2921B**, which is output via the antenna. In

other embodiments, the user's intent is acknowledged on the key fob **292B** via other means, such as via a microphone that accepts audio, a camera that captures images and/or video, or other sensors that are commonly utilized in the art to detect intent from a user including receiving gestures, motion, eye movements, and the like. The data stream may be a 64-bit to 128-bit long signal, which includes one or more of a preamble, a command code, and a rolling code. The signal may be sent at a rate between 2 KHz and 20 KHz, but embodiments are not limited thereto. In response, the receiver **2911B** of the transport **291B** captures the signal from the transmitter **2921B**, demodulates the signal, and sends the data stream to the CPU **2913B**, which decodes the signal and sends commands (e.g., lock the door, unlock the door, etc.) to a command module **2912B**.

[0107] If the key fob **292B** and the transport **291B** use a fixed code between them, replay attacks can be performed. In this case, if the attacker can capture/sniff the fixed code during the short-range communication, the attacker could replay this code to gain entry into the transport **291B**. To improve security, the key fob and the transport **291B** may use a rolling code that changes after each use. Here, the key fob **292B** and the transport **291B** are synchronized with an initial seed **2923B** (e.g., a random number, pseudo-random number, etc.) This is referred to as pairing. The key fob **292B** and the transport **291B** also include a shared algorithm for modifying the initial seed **2914B** each time the button **293B** is pressed. The following keypress will take the result of the previous keypress as an input and transform it into the next number in the sequence. In some cases, the transport **291B** may store multiple next codes (e.g., **255** next codes) in case the keypress on the key fob **292B** is not detected by the transport **291B**. Thus, a number of keypress on the key fob **292B** that are unheard by the transport **291B** do not prevent the transport from becoming out of sync.

[0108] In addition to rolling codes, the key fob **292B** and the transport **291B** may employ other methods to make attacks even more difficult. For example, different frequencies may be used for transmitting the rolling codes. As another example, two-way communication between the transmitter **2921B** and the receiver **2911B** may be used to establish a secure session. As another example, codes may have limited expirations or timeouts. Further, the instant solution as described and depicted with respect to FIG. 2J can be utilized in this and other networks and/or systems, including those that are described and depicted herein.

[0109] FIG. 2K illustrates a controller area network (CAN) **290C** within a transport, according to example embodiments. Referring to FIG. 2K, the CAN **290C** includes a CAN bus **297C** with a high and low terminal and a plurality of electronic control units (ECUs) **291C**, **292C**, **293C**, etc. which are connected to the CAN bus **297C** via wired connections. The CAN bus **297C** is designed to allow microcontrollers and devices to communicate with each other in an application without a host computer. The CAN bus **297C** implements a message-based protocol (i.e., ISO 11898 standards) that allows ECUs **291C-293C** to send commands to one another at a root level. Meanwhile, the ECUs **291C-293C** represent controllers for controlling electrical systems or subsystems within the transport. Examples of the electrical systems include power steering, anti-lock brakes, air-conditioning, tire pressure monitoring, cruise control, and many other features.

[0110] In this example, the ECU **291C** includes a transceiver **2911C** and a microcontroller **2912C**. The transceiver may be used to transmit and receive messages to and from the CAN bus **297C**. For example, the transceiver **2911C** may convert the data from the microcontroller **2912C** into a format of the CAN bus **297C** and also convert data from the CAN bus **297C** into a format for the microcontroller **2912C**. Meanwhile, the microcontroller **2912C** interprets the messages and also decide what messages to send using ECU software installed therein in one example.

[0111] To protect the CAN **290C** from cyber threats, various security protocols may be implemented. For example, sub-networks (e.g., sub-networks A and B, etc.) may be used to divide the CAN **290C** into smaller sub-CANs and limit an attacker's capabilities to access the transport remotely. In the example of FIG. 2K, ECUs **291C** and **292C** may be part of a same sub-network, while ECU **293C** is part of an independent sub-network. Furthermore, a firewall **294C** (or gateway, etc.) may be added to block messages from crossing the CAN bus **297C** across sub-networks. If an attacker gains access to one sub-network, the attacker will not have access to the entire network. To make sub-networks even more secure, the most critical ECUs are not placed on the same sub-network, in one example.

[0112] Although not shown in FIG. 2K, other examples of security controls within a CAN include an intrusion detection system (IDS) which can be added to each sub-network and read all data passing to detect malicious messages. If a malicious message is detected, the IDS can notify the automobile user. Other possible security protocols include encryption/security keys that can be used to obscure messages. As another example, authentication protocols are implemented that enables a message to authenticate itself, in one example.

[0113] In addition to protecting a transport's internal network, transports may also be protected when communicating with external networks such as the Internet. One of the benefits of having a transport connection to a data source such as the Internet is that information from the transport can be sent through a network to remote locations for analysis. Examples of transport information include GPS, onboard diagnostics, tire pressure, and the like. These communication systems are often referred to as telematics because they involve the combination of telecommunications and informatics. Further, the instant solution as described and depicted with respect to FIG. 2K can be utilized in this and other networks and/or systems, including those that are described and depicted herein.

[0114] FIG. 2L illustrates a secure end-to-end transport communication channel according to example embodiments. Referring to FIG. 2L, a telematics network **290D** includes a transport **291D** and a host server **295D** that is disposed at a remote location (e.g., a web server, a cloud platform, a database, etc.) and connected to the transport **291D** via a network such as the Internet. In this example, a device **296D** associated with the host server **295D** may be installed within the network inside the transport **291D**. Furthermore, although not shown, the device **296D** may connect to other elements of the transport **291D**, such as the CAN bus, an onboard diagnostics (OBDII) port, a GPS system, a SIM card, a modem, and the like. The device **296D** may collect data from any of these systems and transfer the data to the server **295D** via the network.

[0115] Secure management of data begins with the transport 291D. In some embodiments, the device 296D may collect information before, during, and after a trip. The data may include GPS data, travel data, passenger information, diagnostic data, fuel data, speed data, and the like. However, the device 296D may only communicate the collected information back to the host server 295D in response to transport ignition and trip completion. Furthermore, communication may only be initiated by the device 296D and not by the host server 295D. As such, the device 296D will not accept communications initiated by outside sources in one example.

[0116] To perform the communication, the device 296D may establish a secured private network between the device 296D and the host server 295D. Here, the device 296D may include a tamper-proof SIM card that provides secure access to a carrier network 294D via a radio tower 292D. When preparing to transmit data to the host server 295D, the device 296D may establish a one-way secure connection with the host server 295D. The carrier network 294D may communicate with the host server 295D using one or more security protocols. As a non-limiting example, the carrier network 294D may communicate with the host server 295D via a VPN tunnel which allows access through a firewall 293D of the host server 295D. As another example, the carrier network 294D may use data encryption (e.g., AES encryption, etc.) when transmitting data to the host server 295D. In some cases, the system may use multiple security measures such as both a VPN and encryption to further secure the data.

[0117] In addition to communicating with external servers, transports may also communicate with each other. In particular, transport-to-transport (V2V) communication systems enable transports to communicate with each other, roadside infrastructures (e.g., traffic lights, signs, cameras, parking meters, etc.), and the like, over a wireless network. The wireless network may include one or more of Wi-Fi networks, cellular networks, dedicated short-range communication (DSRC) networks, and the like. Transports may use V2V communication to provide other transports with information about a transport's speed, acceleration, braking, and direction, to name a few. Accordingly, transports can receive insight into the conditions ahead before such conditions become visible, thus greatly reducing collisions. Further, the instant solution as described and depicted with respect to FIG. 2L can be utilized in this and other networks and/or systems, including those that are described and depicted herein.

[0118] FIG. 2M illustrates an example 290E of transports 293E and 292E performing secured V2V communications using security certificates, according to example embodiments. Referring to FIG. 2M, the transports 293E and 292E may communicate via V2V communications over a short-range network, a cellular network, or the like. Before sending messages, the transports 293E and 292E may sign the messages using a respective public key certificate. For example, the transport 293E may sign a V2V message using a public key certificate 294E. Likewise, the transport 292E may sign a V2V message using a public key certificate 295E. The public key certificates 294E and 295E are associated with the transports 293E and 292E, respectively, in one example.

[0119] Upon receiving the communications from each other, the transports may verify the signatures with a certificate authority 291E or the like. For example, the transport

292E may verify with the certificate authority 291E that the public key certificate 294E used by transport 293E to sign a V2V communication is authentic. If the transport 292E successfully verifies the public key certificate 294E, the transport knows that the data is from a legitimate source. Likewise, the transport 293E may verify with the certificate authority 291E that the public key certificate 295E used by the transport 292E to sign a V2V communication is authentic. Further, the instant solution as described and depicted with respect to FIG. 2M can be utilized in this and other networks and/or systems including those that are described and depicted herein.

[0120] FIG. 2N illustrates yet a further diagram 290F depicting an example of a transport interacting with a security processor and a wireless device, according to example embodiments. In some embodiments, the computer 224 shown in FIG. 2B may include security processor 292F as shown in the process 290F of the example of FIG. 2N. In particular, the security processor 292F may perform authorization, authentication, cryptography (e.g., encryption), and the like, for data transmissions that are sent between ECUs and other devices on a CAN bus of a vehicle, and also data messages that are transmitted between different vehicles.

[0121] In the example of FIG. 2N, the security processor 292F may include an authorization module 293F, an authentication module 294F, and a cryptography module 295F. The security processor 292F may be implemented within the transport's computer and may communicate with other transport elements, for example, the ECUs/CAN network 296F, wired and wireless devices 298F such as wireless network interfaces, input ports, and the like. The security processor 292F may ensure that data frames (e.g., CAN frames, etc.) that are transmitted internally within a transport (e.g., via the ECUs/CAN network 296F) are secure. Likewise, the security processor 292F can ensure that messages transmitted between different transports and devices attached or connected via a wire to the transport's computer are also secured.

[0122] For example, the authorization module 293F may store passwords, usernames, PIN codes, biometric scans, and the like for different transport users. The authorization module 293F may determine whether a user (or technician) has permission to access certain settings such as a transport's computer. In some embodiments, the authorization module may communicate with a network interface to download any necessary authorization information from an external server. When a user desires to make changes to the transport settings or modify technical details of the transport via a console or GUI within the transport or via an attached/connected device, the authorization module 293F may require the user to verify themselves in some way before such settings are changed. For example, the authorization module 293F may require a username, a password, a PIN code, a biometric scan, a predefined line drawing or gesture, and the like. In response, the authorization module 293F may determine whether the user has the necessary permissions (access, etc.) being requested.

[0123] The authentication module 294F may be used to authenticate internal communications between ECUs on the CAN network of the vehicle. As an example, the authentication module 294F may provide information for authenticating communications between the ECUs. As an example, the authentication module 294F may transmit a bit signature algorithm to the ECUs of the CAN network. The ECUs may

use the bit signature algorithm to insert authentication bits into the CAN fields of the CAN frame. All ECUs on the CAN network typically receive each CAN frame. The bit signature algorithm may dynamically change the position, amount, etc., of authentication bits each time a new CAN frame is generated by one of the ECUs. The authentication module 294F may also provide a list of ECUs that are exempt (safe list) and that do not need to use the authentication bits. The authentication module 294F may communicate with a remote server to retrieve updates to the bit signature algorithm and the like.

[0124] The encryption module 295F may store asymmetric key pairs to be used by the transport to communicate with other external user devices and transports. For example, the encryption module 295F may provide a private key to be used by the transport to encrypt/decrypt communications, while the corresponding public key may be provided to other user devices and transports to enable the other devices to decrypt/encrypt the communications. The encryption module 295F may communicate with a remote server to receive new keys, updates to keys, keys of new transports, users, etc., and the like. The encryption module 295F may also transmit any updates to a local private/public key pair to the remote server.

[0125] FIG. 3A illustrates a flow diagram 300, according to example embodiments. Referring to FIG. 3A, the current solution includes one or more of determining an inability for an entity to provide electricity over a threshold to an area 302; determining available energy from a group of electric vehicles from locations currently receiving energy in the area and a group of energy storage units in the area greater than the threshold 304; providing the determined available energy to locations not currently receiving energy from the entity 306.

[0126] FIG. 3B illustrates another flow diagram 320, according to example embodiments. Referring to FIG. 3B, the current solution performs one or more of wherein the threshold relates to an amount of electricity required to power a first location in the area 322; wherein the threshold relates to a cost of electricity associated with powering a first location in the area 323; wherein determining available energy from the group of electric vehicles and the group of energy storage units includes predicting an anticipated surplus of available energy 324; wherein determining available energy from the group of electric vehicles and the group of energy storage units includes energy expended traveling between the locations not currently receiving energy from the entity and the locations of the group of vehicles and the group of energy storage units 325; wherein providing the determined available energy includes dispatching a vehicle from the group of electric vehicles to locations not currently receiving energy 326; wherein providing the determined available energy includes dispatching a vehicle from locations not currently receiving energy to the group of energy storage units 327.

[0127] FIG. 3C illustrates yet another flow diagram 340, according to example embodiments. Referring to FIG. 3C, the flow diagram includes one or more of receiving a confirmation of an event from one or more elements described or depicted herein, wherein the confirmation comprises a blockchain consensus between peers represented by any of the elements 342 and executing a smart contract to record the confirmation on a blockchain-based on the blockchain consensus 344.

[0128] FIG. 4 illustrates a machine learning transport network diagram 400, according to example embodiments. The network 400 includes a transport 402 that interfaces with a machine learning subsystem 406. The transport includes one or more sensors 404.

[0129] The machine learning subsystem 406 contains a learning model 408, which is an artifact created by a machine learning training system 410 that generates predictions by finding patterns in one or more training data sets. In some embodiments, the machine learning subsystem 406 resides in the transport node 402. An artifact is used to describe an output created by a training process, such as a checkpoint, a file, or a model. In other embodiments, the machine learning subsystem 406 resides outside of the transport node 402.

[0130] The transport 402 sends data from the one or more sensors 404 to the machine learning subsystem 406. The machine learning subsystem 406 provides the one or more sensor 404 data to the learning model 408, which returns one or more predictions. The machine learning subsystem 406 sends one or more instructions to the transport 402 based on the predictions from the learning model 408.

[0131] In a further embodiment, the transport 402 may send the one or more sensor 404 data to the machine learning training system 410. In yet another example, the machine learning subsystem 406 may send the sensor 404 data to the machine learning subsystem 410. One or more of the applications, features, steps, solutions, etc., described and/or depicted herein may utilize the machine learning network 400 as described herein.

[0132] FIG. 5A illustrates an example vehicle configuration 500 for managing database transactions associated with a vehicle, according to example embodiments. Referring to FIG. 5A, as a particular transport/vehicle 525 is engaged in transactions (e.g., vehicle service, dealer transactions, delivery/pickup, transportation services, etc.), the vehicle may receive assets 510 and/or expel/transfer assets 512 according to a transaction(s). A transport processor 526 resides in the vehicle 525 and communication exists between the transport processor 526, a database 530, a transport processor 526 and the transaction module 520. The transaction module 520 may record information, such as assets, parties, credits, service descriptions, date, time, location, results, notifications, unexpected events, etc. Those transactions in the transaction module 520 may be replicated into a database 530. The database 530 can be one of a SQL database, an RDBMS, a relational database, a non-relational database, a blockchain, a distributed ledger, and may be on board the transport, may be off-board the transport, may be accessed directly and/or through a network, or be accessible to the transport.

[0133] FIG. 5B illustrates an example vehicle configuration 550 for managing database transactions conducted among various vehicles, according to example embodiments. The vehicle 525 may engage with another vehicle 508 to perform various actions such as to share, transfer, acquire service calls, etc. when the vehicle has reached a status where the services need to be shared with another vehicle. For example, the vehicle 508 may be due for a battery charge and/or may have an issue with a tire and may be in route to pick up a package for delivery. A transport processor 528 resides in the vehicle 508 and communication exists between the transport processor 528, a database 554, and the transaction module 552. The vehicle 508 may notify

another vehicle **525**, which is in its network and which operates on its blockchain member service. A transport processor **526** resides in the vehicle **525** and communication exists between the transport processor **526**, a database **530**, the transport processor **526** and a transaction module **520**. The vehicle **525** may then receive the information via a wireless communication request to perform the package pickup from the vehicle **508** and/or from a server (not shown). The transactions are logged in the transaction modules **552** and **520** of both vehicles. The credits are transferred from vehicle **508** to vehicle **525** and the record of the transferred service is logged in the database **530/554** assuming that the blockchains are different from one another, or are logged in the same blockchain used by all members. The database **554** can be one of a SQL database, an RDBMS, a relational database, a non-relational database, a blockchain, a distributed ledger, and may be on board the transport, may be off-board the transport, may be accessible directly and/or through a network.

[0134] FIG. 6A illustrates a blockchain architecture configuration **600**, according to example embodiments. Referring to FIG. 6A, the blockchain architecture **600** may include certain blockchain elements, for example, a group of blockchain member nodes **602-606** as part of a blockchain group **610**. In one example embodiment, a permissioned blockchain is not accessible to all parties but only to those members with permissioned access to the blockchain data. The blockchain nodes participate in a number of activities, such as blockchain entry addition and validation process (consensus). One or more of the blockchain nodes may endorse entries based on an endorsement policy and may provide an ordering service for all blockchain nodes. A blockchain node may initiate a blockchain action (such as an authentication) and seek to write to a blockchain immutable ledger stored in the blockchain, a copy of which may also be stored on the underpinning physical infrastructure.

[0135] The blockchain transactions **620** are stored in memory of computers as the transactions are received and approved by the consensus model dictated by the members' nodes. Approved transactions **626** are stored in current blocks of the blockchain and committed to the blockchain via a committal procedure, which includes performing a hash of the data contents of the transactions in a current block and referencing a previous hash of a previous block. Within the blockchain, one or more smart contracts **630** may exist that define the terms of transaction agreements and actions included in smart contract executable application code **632**, such as registered recipients, vehicle features, requirements, permissions, sensor thresholds, etc. The code may be configured to identify whether requesting entities are registered to receive vehicle services, what service features they are entitled/required to receive given their profile statuses and whether to monitor their actions in subsequent events. For example, when a service event occurs and a user is riding in the vehicle, the sensor data monitoring may be triggered, and a certain parameter, such as a vehicle charge level, may be identified as being above/below a particular threshold for a particular period of time, then the result may be a change to a current status, which requires an alert to be sent to the managing party (i.e., vehicle owner, vehicle operator, server, etc.) so the service can be identified and stored for reference. The vehicle sensor data collected may be based on types of sensor data used to collect information about vehicle's status. The sensor data may also be the basis

for the vehicle event data **634**, such as a location(s) to be traveled, an average speed, a top speed, acceleration rates, whether there were any collisions, was the expected route taken, what is the next destination, whether safety measures are in place, whether the vehicle has enough charge/fuel, etc. All such information may be the basis of smart contract terms **630**, which are then stored in a blockchain. For example, sensor thresholds stored in the smart contract can be used as the basis for whether a detected service is necessary and when and where the service should be performed.

[0136] FIG. 6B illustrates a shared ledger configuration, according to example embodiments. Referring to FIG. 6B, the blockchain logic example **640** includes a blockchain application interface **642** as an API or plug-in application that links to the computing device and execution platform for a particular transaction. The blockchain configuration **640** may include one or more applications, which are linked to application programming interfaces (APIs) to access and execute stored program/application code (e.g., smart contract executable code, smart contracts, etc.), which can be created according to a customized configuration sought by participants and can maintain their own state, control their own assets, and receive external information. This can be deployed as an entry and installed, via appending to the distributed ledger, on all blockchain nodes.

[0137] The smart contract application code **644** provides a basis for the blockchain transactions by establishing application code, which when executed causes the transaction terms and conditions to become active. The smart contract **630**, when executed, causes certain approved transactions **626** to be generated, which are then forwarded to the blockchain platform **652**. The platform includes a security/authorization **658**, computing devices, which execute the transaction management **656** and a storage portion **654** as a memory that stores transactions and smart contracts in the blockchain.

[0138] The blockchain platform may include various layers of blockchain data, services (e.g., cryptographic trust services, virtual execution environment, etc.), and underpinning physical computer infrastructure that may be used to receive and store new entries and provide access to auditors, which are seeking to access data entries. The blockchain may expose an interface that provides access to the virtual execution environment necessary to process the program code and engage the physical infrastructure. Cryptographic trust services may be used to verify entries such as asset exchange entries and keep information private.

[0139] The blockchain architecture configuration of FIGS. 6A and 6B may process and execute program/application code via one or more interfaces exposed, and services provided, by the blockchain platform. As a non-limiting example, smart contracts may be created to execute reminders, updates, and/or other notifications subject to the changes, updates, etc. The smart contracts can themselves be used to identify rules associated with authorization and access requirements and usage of the ledger. For example, the information may include a new entry, which may be processed by one or more processing entities (e.g., processors, virtual machines, etc.) included in the blockchain layer. The result may include a decision to reject or approve the new entry based on the criteria defined in the smart contract

and/or a consensus of the peers. The physical infrastructure may be utilized to retrieve any of the data or information described herein.

[0140] Within smart contract executable code, a smart contract may be created via a high-level application and programming language, and then written to a block in the blockchain. The smart contract may include executable code that is registered, stored, and/or replicated with a blockchain (e.g., distributed network of blockchain peers). An entry is an execution of the smart contract code, which can be performed in response to conditions associated with the smart contract being satisfied. The executing of the smart contract may trigger a trusted modification(s) to a state of a digital blockchain ledger. The modification(s) to the blockchain ledger caused by the smart contract execution may be automatically replicated throughout the distributed network of blockchain peers through one or more consensus protocols.

[0141] The smart contract may write data to the blockchain in the format of key-value pairs. Furthermore, the smart contract code can read the values stored in a blockchain and use them in application operations. The smart contract code can write the output of various logic operations into the blockchain. The code may be used to create a temporary data structure in a virtual machine or other computing platform. Data written to the blockchain can be public and/or can be encrypted and maintained as private. The temporary data that is used/generated by the smart contract is held in memory by the supplied execution environment, then deleted once the data needed for the blockchain is identified.

[0142] A smart contract executable code may include the code interpretation of a smart contract, with additional features. As described herein, the smart contract executable code may be program code deployed on a computing network, where it is executed and validated by chain validators together during a consensus process. The smart contract executable code receives a hash and retrieves from the blockchain a hash associated with the data template created by use of a previously stored feature extractor. If the hashes of the hash identifier and the hash created from the stored identifier template data match, then the smart contract executable code sends an authorization key to the requested service. The smart contract executable code may write to the blockchain data associated with the cryptographic details.

[0143] FIG. 6C illustrates a blockchain configuration for storing blockchain transaction data, according to example embodiments. Referring to FIG. 6C, the example configuration 660 provides for the vehicle 662, the user device 664 and a server 666 sharing information with a distributed ledger (i.e., blockchain) 668. The server may represent a service provider entity inquiring with a vehicle service provider to share user profile rating information in the event that a known and established user profile is attempting to rent a vehicle with an established rated profile. The server 666 may be receiving and processing data related to a vehicle's service requirements. As the service events occur, such as the vehicle sensor data indicates a need for fuel/charge, a maintenance service, etc., a smart contract may be used to invoke rules, thresholds, sensor information gathering, etc., which may be used to invoke the vehicle service event. The blockchain transaction data 670 is saved for each transaction, such as the access event, the subsequent updates to a vehicle's service status, event updates, etc. The trans-

actions may include the parties, the requirements (e.g., 18 years of age, service eligible candidate, valid driver's license, etc.), compensation levels, the distance traveled during the event, the registered recipients permitted to access the event and host a vehicle service, rights/permissions, sensor data retrieved during the vehicle event operation to log details of the next service event and identify a vehicle's condition status, and thresholds used to make determinations about whether the service event was completed and whether the vehicle's condition status has changed.

[0144] FIG. 6D illustrates blockchain blocks 680 that can be added to a distributed ledger, according to example embodiments, and contents of block structures 682A to 682n. Referring to FIG. 6D, clients (not shown) may submit entries to blockchain nodes to enact activity on the blockchain. As an example, clients may be applications that act on behalf of a requester, such as a device, person or entity to propose entries for the blockchain. The plurality of blockchain peers (e.g., blockchain nodes) may maintain a state of the blockchain network and a copy of the distributed ledger. Different types of blockchain nodes/peers may be present in the blockchain network including endorsing peers, which simulate and endorse entries proposed by clients and committing peers which verify endorsements, validate entries, and commit entries to the distributed ledger. In this example, the blockchain nodes may perform the role of endorser node, committer node, or both.

[0145] The instant system includes a blockchain that stores immutable, sequenced records in blocks, and a state database (current world state) maintaining a current state of the blockchain. One distributed ledger may exist per channel and each peer maintains its own copy of the distributed ledger for each channel of which they are a member. The instant blockchain is an entry log, structured as hash-linked blocks where each block contains a sequence of N entries. Blocks may include various components such as those shown in FIG. 6D. The linking of the blocks may be generated by adding a hash of a prior block's header within a block header of a current block. In this way, all entries on the blockchain are sequenced and cryptographically linked together preventing tampering with blockchain data without breaking the hash links. Furthermore, because of the links, the latest block in the blockchain represents every entry that has come before it. The instant blockchain may be stored on a peer file system (local or attached storage), which supports an append-only blockchain workload.

[0146] The current state of the blockchain and the distributed ledger may be stored in the state database. Here, the current state data represents the latest values for all keys ever included in the chain entry log of the blockchain. Smart contract executable code invocations execute entries against the current state in the state database. To make these smart contract executable code interactions extremely efficient, the latest values of all keys are stored in the state database. The state database may include an indexed view into the entry log of the blockchain, it can therefore be regenerated from the chain at any time. The state database may automatically get recovered (or generated if needed) upon peer startup, before entries are accepted.

[0147] Endorsing nodes receive entries from clients and endorse the entry based on simulated results. Endorsing nodes hold smart contracts, which simulate the entry proposals. When an endorsing node endorses an entry, the

endorsing nodes creates an entry endorsement, which is a signed response from the endorsing node to the client application indicating the endorsement of the simulated entry. The method of endorsing an entry depends on an endorsement policy that may be specified within smart contract executable code. An example of an endorsement policy is “the majority of endorsing peers must endorse the entry.” Different channels may have different endorsement policies. Endorsed entries are forward by the client application to an ordering service.

[0148] The ordering service accepts endorsed entries, orders them into a block, and delivers the blocks to the committing peers. For example, the ordering service may initiate a new block when a threshold of entries has been reached, a timer times out, or another condition. In this example, blockchain node is a committing peer that has received a data block 682A for storage on the blockchain. The ordering service may be made up of a cluster of orderers. The ordering service does not process entries, smart contracts, or maintain the shared ledger. Rather, the ordering service may accept the endorsed entries and specifies the order in which those entries are committed to the distributed ledger. The architecture of the blockchain network may be designed such that the specific implementation of ‘ordering’ (e.g., Solo, Kafka, BFT, etc.) becomes a pluggable component.

[0149] Entries are written to the distributed ledger in a consistent order. The order of entries is established to ensure that the updates to the state database are valid when they are committed to the network. Unlike a cryptocurrency blockchain system (e.g., Bitcoin, etc.) where ordering occurs through the solving of a cryptographic puzzle, or mining, in this example the parties of the distributed ledger may choose the ordering mechanism that best suits that network.

[0150] Referring to FIG. 6D, a block 682A (also referred to as a data block) that is stored on the blockchain and/or the distributed ledger may include multiple data segments such as a block header 684A to 684n, transaction-specific data 686A to 686n, and block metadata 688A to 688/1. It should be appreciated that the various depicted blocks and their contents, such as block 682A and its contents are merely for purposes of an example and are not meant to limit the scope of the example embodiments. In some cases, both the block header 684A and the block metadata 688A may be smaller than the transaction-specific data 686A, which stores entry data; however, this is not a requirement. The block 682A may store transactional information of N entries (e.g., 100, 500, 1000, 2000, 3000, etc.) within the block data 690A to 690n. The block 682A may also include a link to a previous block (e.g., on the blockchain) within the block header 684A. In particular, the block header 684A may include a hash of a previous block’s header. The block header 684A may also include a unique block number, a hash of the block data 690A of the current block 682A, and the like. The block number of the block 682A may be unique and assigned in an incremental/sequential order starting from zero. The first block in the blockchain may be referred to as a genesis block, which includes information about the blockchain, its members, the data stored therein, etc.

[0151] The block data 690A may store entry information of each entry that is recorded within the block. For example, the entry data may include one or more of a type of the entry, a version, a timestamp, a channel ID of the distributed ledger, an entry ID, an epoch, a payload visibility, a smart

contract executable code path (deploy tx), a smart contract executable code name, a smart contract executable code version, input (smart contract executable code and functions), a client (creator) identify such as a public key and certificate, a signature of the client, identities of endorsers, endorser signatures, a proposal hash, smart contract executable code events, response status, namespace, a read set (list of key and version read by the entry, etc.), a write set (list of key and value, etc.), a start key, an end key, a list of keys, a Merkel tree query summary, and the like. The entry data may be stored for each of the N entries.

[0152] In some embodiments, the block data 690A may also store transaction-specific data 686A, which adds additional information to the hash-linked chain of blocks in the blockchain. Accordingly, the data 686A can be stored in an immutable log of blocks on the distributed ledger. Some of the benefits of storing such data 686A are reflected in the various embodiments disclosed and depicted herein. The block metadata 688A may store multiple fields of metadata (e.g., as a byte array, etc.). Metadata fields may include signature on block creation, a reference to a last configuration block, an entry filter identifying valid and invalid entries within the block, last offset persisted of an ordering service that ordered the block, and the like. The signature, the last configuration block, and the orderer metadata may be added by the ordering service. Meanwhile, a committer of the block (such as a blockchain node) may add validity/invalidity information based on an endorsement policy, verification of read/write sets, and the like. The entry filter may include a byte array of a size equal to the number of entries in the block data 610A and a validation code identifying whether an entry was valid/invalid.

[0153] The other blocks 682B to 682n in the blockchain also have headers, files, and values. However, unlike the first block 682A, each of the headers 684A to 684n in the other blocks includes the hash value of an immediately preceding block. The hash value of the immediately preceding block may be just the hash of the header of the previous block or may be the hash value of the entire previous block. By including the hash value of a preceding block in each of the remaining blocks, a trace can be performed from the Nth block back to the genesis block (and the associated original file) on a block-by-block basis, as indicated by arrows 692, to establish an auditable and immutable chain-of-custody.

[0154] The above embodiments may be implemented in hardware, in a computer program executed by a processor, in firmware, or in a combination of the above. A computer program may be embodied on a computer readable medium, such as a storage medium. For example, a computer program may reside in random access memory (“RAM”), flash memory, read-only memory (“ROM”), erasable programmable read-only memory (“EPROM”), electrically erasable programmable read-only memory (“EEPROM”), registers, hard disk, a removable disk, a compact disk read-only memory (“CD-ROM”), or any other form of storage medium known in the art.

[0155] An exemplary storage medium may be coupled to the processor such that the processor may read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside in an application-specific integrated circuit (“ASIC”). In the alternative, the processor and the storage medium may reside as discrete components. For example, FIG. 7 illus-

trates an example computer system architecture **700**, which may represent or be integrated in any of the above-described components, etc.

[0156] FIG. 7 is not intended to suggest any limitation as to the scope of use or functionality of embodiments of the application described herein. Regardless, the computing node **700** is capable of being implemented and/or performing any of the functionality set forth hereinabove.

[0157] In computing node **700** there is a computer system/server **702**, which is operational with numerous other general purpose or special purpose computing system environments or configurations. Examples of well-known computing systems, environments, and/or configurations that may be suitable for use with computer system/server **702** include, but are not limited to, personal computer systems, server computer systems, thin clients, thick clients, hand-held or laptop devices, multiprocessor systems, microprocessor-based systems, set-top boxes, programmable consumer electronics, network PCs, minicomputer systems, mainframe computer systems, and distributed cloud computing environments that include any of the above systems or devices, and the like.

[0158] Computer system/server **702** may be described in the general context of computer system-executable instructions, such as program modules, being executed by a computer system. Generally, program modules may include routines, programs, objects, components, logic, data structures, and so on that perform particular tasks or implement particular abstract data types. Computer system/server **702** may be practiced in distributed cloud computing environments where tasks are performed by remote processing devices that are linked through a communications network. In a distributed cloud computing environment, program modules may be located in both local and remote computer system storage media including memory storage devices.

[0159] As shown in FIG. 7, computer system/server **702** in cloud computing node **700** is shown in the form of a general-purpose computing device. The components of computer system/server **702** may include, but are not limited to, one or more processors or processing units **704**, a system memory **706**, and a bus that couples various system components including system memory **706** to processor **704**.

[0160] The bus represents one or more of any of several types of bus structures, including a memory bus or memory controller, a peripheral bus, an accelerated graphics port, and a processor or local bus using any of a variety of bus architectures. By way of example, and not limitation, such architectures include Industry Standard Architecture (ISA) bus, Micro Channel Architecture (MCA) bus, Enhanced ISA (EISA) bus, Video Electronics Standards Association (VESA) local bus, and Peripheral Component Interconnects (PCI) bus.

[0161] Computer system/server **702** typically includes a variety of computer system readable media. Such media may be any available media that is accessible by computer system/server **702**, and it includes both volatile and non-volatile media, removable and non-removable media. System memory **706**, in one example, implements the flow diagrams of the other figures. The system memory **706** can include computer system readable media in the form of volatile memory, such as random-access memory (RAM) **708** and/or cache memory **710**. Computer system/server **702** may further include other removable/non-removable, volatile/non-volatile computer system storage media. By way of

example only, memory **706** can be provided for reading from and writing to a non-removable, non-volatile magnetic media (not shown and typically called a “hard drive”). Although not shown, a magnetic disk drive for reading from and writing to a removable, non-volatile magnetic disk (e.g., a “floppy disk”), and an optical disk drive for reading from or writing to a removable, non-volatile optical disk such as a CD-ROM, DVD-ROM or other optical media can be provided. In such instances, each can be connected to the bus by one or more data media interfaces. As will be further depicted and described below, memory **706** may include at least one program product having a set (e.g., at least one) of program modules that are configured to carry out the functions of various embodiments of the application.

[0162] Program/utility, having a set (at least one) of program modules, may be stored in memory **706** by way of example, and not limitation, as well as an operating system, one or more application programs, other program modules, and program data. Each of the operating system, one or more application programs, other program modules, and program data or some combination thereof, may include an implementation of a networking environment. Program modules generally carry out the functions and/or methodologies of various embodiments of the application as described herein.

[0163] As will be appreciated by one skilled in the art, aspects of the present application may be embodied as a system, method, or computer program product. Accordingly, aspects of the present application may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a “circuit,” “module” or “system.” Furthermore, aspects of the present application may take the form of a computer program product embodied in one or more computer readable medium(s) having computer readable program code embodied thereon.

[0164] Computer system/server **702** may also communicate with one or more external devices via an I/O device **712** (such as an I/O adapter), which may include a keyboard, a pointing device, a display, a voice recognition module, etc., one or more devices that enable a user to interact with computer system/server **702**, and/or any devices (e.g., network card, modem, etc.) that enable computer system/server **702** to communicate with one or more other computing devices. Such communication can occur via I/O interfaces of the device **712**. Still yet, computer system/server **702** can communicate with one or more networks such as a local area network (LAN), a general wide area network (WAN), and/or a public network (e.g., the Internet) via a network adapter. As depicted, device **712** communicates with the other components of computer system/server **702** via a bus. It should be understood that although not shown, other hardware and/or software components could be used in conjunction with computer system/server **702**. Examples, include, but are not limited to: microcode, device drivers, redundant processing units, external disk drive arrays, RAID systems, tape drives, and data archival storage systems, etc.

[0165] Although an exemplary embodiment of at least one of a system, method, and non-transitory computer readable medium has been illustrated in the accompanied drawings and described in the foregoing detailed description, it will be understood that the application is not limited to the embodiments disclosed, but is capable of numerous rearrangements,

modifications, and substitutions as set forth and defined by the following claims. For example, the capabilities of the system of the various figures can be performed by one or more of the modules or components described herein or in a distributed architecture and may include a transmitter, receiver or pair of both. For example, all or part of the functionality performed by the individual modules, may be performed by one or more of these modules. Further, the functionality described herein may be performed at various times and in relation to various events, internal or external to the modules or components. Also, the information sent between various modules can be sent between the modules via at least one of: a data network, the Internet, a voice network, an Internet Protocol network, a wireless device, a wired device and/or via plurality of protocols. Also, the messages sent or received by any of the modules may be sent or received directly and/or via one or more of the other modules.

[0166] One skilled in the art will appreciate that a “system” could be embodied as a personal computer, a server, a console, a personal digital assistant (PDA), a cell phone, a tablet computing device, a smartphone or any other suitable computing device, or combination of devices. Presenting the above-described functions as being performed by a “system” is not intended to limit the scope of the present application in any way but is intended to provide one example of many embodiments. Indeed, methods, systems and apparatuses disclosed herein may be implemented in localized and distributed forms consistent with computing technology.

[0167] It should be noted that some of the system features described in this specification have been presented as modules to more particularly emphasize their implementation independence. For example, a module may be implemented as a hardware circuit comprising custom very-large-scale integration (VLSI) circuits or gate arrays, off-the-shelf semiconductors such as logic chips, transistors, or other discrete components. A module may also be implemented in programmable hardware devices such as field-programmable gate arrays, programmable array logic, programmable logic devices, graphics processing units, or the like.

[0168] A module may also be at least partially implemented in software for execution by various types of processors. An identified unit of executable code may, for instance, comprise one or more physical or logical blocks of computer instructions that may, for instance, be organized as an object, procedure, or function. Nevertheless, the executables of an identified module need not be physically located together but may comprise disparate instructions stored in different locations that, when joined logically together, comprise the module and achieve the stated purpose for the module. Further, modules may be stored on a computer-readable medium, which may be, for instance, a hard disk drive, flash device, random access memory (RAM), tape, or any other such medium used to store data.

[0169] Indeed, a module of executable code could be a single instruction, or many instructions, and may even be distributed over several different code segments, among different programs, and across several memory devices. Similarly, operational data may be identified and illustrated herein within modules and may be embodied in any suitable form and organized within any suitable type of data structure. The operational data may be collected as a single data set or may be distributed over different locations, including

over different storage devices, and may exist, at least partially, merely as electronic signals on a system or network.

[0170] It will be readily understood that the components of the application, as generally described and illustrated in the figures herein, may be arranged and designed in a wide variety of different configurations. Thus, the detailed description of the embodiments is not intended to limit the scope of the application as claimed but is merely representative of selected embodiments of the application.

[0171] One having ordinary skill in the art will readily understand that the above may be practiced with steps in a different order and/or with hardware elements in configurations that are different from those which are disclosed. Therefore, although the application has been described based upon these preferred embodiments, it would be apparent to those of skill in the art that certain modifications, variations, and alternative constructions would be apparent.

[0172] While preferred embodiments of the present application have been described, it is to be understood that the embodiments described are illustrative only and the scope of the application is to be defined solely by the appended claims when considered with a full range of equivalents and modifications (e.g., protocols, hardware devices, software platforms etc.) thereto.

What is claimed is:

1. A method, comprising:

determining an inability for an entity to provide electricity over a threshold to an area;

determining available energy from a group of electric vehicles from locations currently receiving energy in the area and a group of energy storage units in the area greater than the threshold;

providing the determined available energy to locations not currently receiving energy from the entity.

2. The method of claim 1, wherein the threshold relates to an amount of electricity required to power a first location in the area.

3. The method of claim 1, wherein the threshold relates to a cost of electricity associated with powering a first location in the area.

4. The method of claim 1, wherein determining available energy from the group of electric vehicles and the group of energy storage units includes predicting an anticipated surplus of available energy.

5. The method of claim 1, wherein determining available energy from the group of electric vehicles and the group of energy storage units includes energy expended traveling between the locations not currently receiving energy from the entity and the locations of the group of vehicles and the group of energy storage units.

6. The method of claim 1, wherein providing the determined available energy includes dispatching a vehicle from the group of electric vehicles to locations not currently receiving energy.

7. The method of claim 1, wherein providing the determined available energy includes dispatching a vehicle from locations not currently receiving energy to the group of energy storage units.

8. A system, comprising:

a memory to store a set of instructions; and

a processor to execute the set of instructions, wherein the processor causes the system to:

determine an inability for an entity to provide electricity over a threshold to an area;

determine available energy from a group of electric vehicles from locations currently receiving energy in the area and a group of energy storage units in the area greater than the threshold;

providing the determined available energy to locations not currently receiving energy from the entity.

9. The system of claim **8**, wherein the threshold relates to an amount of electricity required to power a first location in the area.

10. The system of claim **8**, wherein the threshold relates to a cost of electricity associated with powering a first location in the area.

11. The system of claim **8**, further comprising instructions executable by the processor to cause the system to determine available energy from the group of electric vehicles and the group of energy storage units including predicting an anticipated surplus of available energy.

12. The system of claim **8**, further comprising instructions executable by the processor to cause the system to determine available energy from the group of electric vehicles and the group of energy storage units including energy expended traveling between the locations not currently receiving energy from the entity and the locations of the group of vehicles and the group of energy storage units.

13. The system of claim **8**, further comprising instructions executable by the processor to cause the system to provide the determined available energy including dispatching a vehicle from the group of electric vehicles to locations not currently receiving energy.

14. The system of claim **8**, wherein providing the determined available energy includes dispatching a vehicle from locations not currently receiving energy to the group of energy storage units.

15. A computer readable storage medium comprising instructions, that when read by a processor, cause the processor to perform:

determining an inability for an entity to provide electricity over a threshold to an area;

determining available energy from a group of electric vehicles from locations currently receiving energy in the area and a group of energy storage units in the area greater than the threshold;

providing the determined available energy to locations not currently receiving energy from the entity.

16. The computer readable medium of claim **15**, wherein the threshold relates to an amount of electricity required to power a first location in the area.

17. The computer readable medium of claim **15**, wherein the threshold relates to a cost of electricity associated with powering a first location in the area.

18. The computer readable medium of claim **15**, wherein determining available energy from the group of electric vehicles and the group of energy storage units includes predicting an anticipated surplus of available energy.

19. The computer readable medium of claim **15**, wherein providing the determined available energy includes dispatching a vehicle from the group of electric vehicles to locations not currently receiving energy.

20. The computer readable medium of claim **15**, wherein providing the determined available energy includes dispatching a vehicle from locations not currently receiving energy to the group of energy storage units.

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